Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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Abstract

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization provides insufficient guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance signals, and static objective preferences cannot adapt to different evolutionary phases. To address these limitations, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism. First, to overcome inefficient initialization, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and recovering population diversity when stagnation occurs. Second, to exploit discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and steers elite solutions toward historically effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling that emphasizes early dimensionality reduction and late-stage accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL achieves the best classification performance among all competitors, with a mean accuracy of 0.9574 and only 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm statistically significant improvements over all 13 competitors.

Keywords: feature selection, particle swarm optimization, mutual information, knowledge-guided elite refinement, two-phase weight scheduling

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism that integrates prior knowledge injection, evolutionary knowledge accumulation, and adaptive objective scheduling into the multi-task PSO architecture. The proposed framework consists of three complementary mechanisms that address different aspects of high-dimensional FS optimization.

First, mutual-information-guided initialization with adaptive re-initialization incorporates prior feature-relevance knowledge into population construction, reduces early-stage search waste caused by unguided initialization, and restores population diversity when stagnation occurs. Second, the Knowledge-guided Elite Refinement (KER) strategy accumulates cross-iteration feature-importance knowledge from fitness comparisons performed during evolution, applies exponential decay to reduce the influence of outdated evidence, and guides elite solutions toward historically informative features within a bounded refinement budget. Third, Two-Phase Weight Scheduling

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(TPS) dynamically adjusts the dual-objective preference from feature reduction to classification refinement, matching optimization priorities with the evolving requirements of high-dimensional FS. These mechanisms operate at different levels of the evolutionary process, and their individual contributions can be systematically verified through component contribution analysis.

The main contributions of this paper are summarized as follows:

1. An MI-guided initialization scheme with adaptive re-initialization is proposed to incorporate feature-relevance knowledge into population construction and restore population diversity when evolutionary stagnation occurs.
2. A Knowledge-guided Elite Refinement (KER) strategy is developed to accumulate cross-iteration feature-importance knowledge from evolutionary feedback, reduce outdated information through exponential decay, and guide elite solutions toward historically effective features within a bounded refinement budget.
3. A Two-Phase Weight Scheduling (TPS) mechanism is introduced to replace static objective weighting with a stage-aware optimization strategy, emphasizing aggressive feature reduction in early evolution and accuracy refinement in later evolution.
4. Comprehensive experiments are conducted on 20 high-dimensional benchmark datasets containing 500 to 54,675 features against 14 fixed-length evolutionary FS methods. Statistical tests, ablation studies, and parameter sensitivity analyses are performed to evaluate the effectiveness and robustness of KDS-EL.

Extensive experiments demonstrate that KDS-EL achieves the best classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features. Compared with the strongest single-objective and multi-objective competitors, KDS-EL improves accuracy by 3.97 and 4.42 percentage points, respectively, while achieving a 3.4× speedup over MEL (51.4 s vs. 175.0 s). Furthermore, Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm that the improvements are statistically significant over all 13 competitors.

The remainder of this paper is organized as follows. Section 2 reviews feature selection methods and evolutionary FS algorithms. Section 3 introduces the background knowledge, including PSO, mutual-information-based feature relevance estimation, and the MEL framework. Section 4 presents the proposed KDS-EL framework and describes its three mechanisms in detail. Section 5 reports the experimental configuration and results, including comparative experiments, component contribution analysis, and parameter sensitivity analysis. Finally, Section 6 concludes this paper and discusses future research directions.

2. Related Work

This section provides a systematic survey of evolutionary feature selection paradigms, multi-objective versus single-objective formulations in high-dimensional spaces, evolutionary multi-task optimization, and knowledge-guided meta-heuristics, establishing the theoretical demarcation and unique positioning of KDS-EL.

2.1. Evolutionary Swarm Intelligence for Feature Selection

Feature selection is a combinatorial NP-hard optimization problem that aims to identify an optimal subset of $d \ll D$ relevant features, maximizing predictive accuracy while eliminating redundant and uninformative dimensions [1, 2]. Traditional feature selection techniques are categorized into three classical paradigms:

- **Filter Methods**: Filter techniques evaluate feature subsets based on intrinsic statistical properties, such as mutual information (mRMR [13]), correlation (FCBF [14]), or graph-theoretic centrality [15]. Although computationally efficient ($O(D \log D)$), filter methods evaluate features independently or pairwise, fundamentally ignoring complex multivariate interactions and resulting in suboptimal accuracy when applied to non-linear classification landscapes.

- **Wrapper Methods**: Wrapper approaches evaluate candidate subsets using the direct predictive feedback of a machine learning classifier [4]. While wrappers capture intricate feature dependencies and deliver superior classification performance, exhaustive evaluation is intractable in high-dimensional spaces (2^D combinatorial states), necessitating stochastic global search heuristics.

- **Embedded Methods**: Embedded methods incorporate feature selection directly into model training via regularization, such as LASSO/Bolasso [16] and tree-based ensemble importance (FS-GBDT [17]). However, embedded rankings are tightly coupled with specific classifier inductive biases and frequently struggle to generalize across diverse classification architectures.

To harness the predictive fidelity of wrappers while mitigating combinatorial computational bottlenecks, Swarm Intelligence (SI) and Evolutionary Computation (EC) have emerged as the dominant paradigms for feature selection [3**?**]. Classical meta-heuristics, including Binary Particle Swarm Optimization (BPSO) [7], Genetic Algorithms (GA) [18], Ant Colony Optimization (ACO) [19], and Artificial Bee Colony (ABC) [20], navigate high-dimensional spaces via population-based stochastic exploration.

To combat the curse of dimensionality in small-sample regimes ($p \gg n$), numerous specialized single-objective evolutionary algorithms have been developed: self-adaptive swarm mechanisms [21], progressively refined search trajectories [22], self-adaptive differential evolution with adaptive weighting (SaWDE) [9], multi-layer perceptron-guided PSO (IGPSO) [10], multi-criterion scoring PSO (PSO-CSM) [11], reinforcement-learning-assisted swarm control [23], and divide-and-conquer subspace decomposition [24]. While these methods enhance local search operators, they universally treat feature perturbation as a memoryless stochastic trial, discarding the accumulated evolutionary feedback across generations.

2.2. Multi-Objective versus Single-Objective Formulations at Scale

Feature selection naturally involves two conflicting objectives: minimizing classification error and minimizing the number of selected features. Multi-Objective Feature Selection (MOFS) algorithms model this trade-off by searching for a Pareto front of non-dominated solutions. Pioneered by Xue et al. [25], diverse multi-objective evolutionary algorithms have been adapted to feature selection, including NSGA-II-FS [26], MOEA/D-FS [27], MOPSO-FS [28], and hybrid filter-wrapper multi-objective architectures [29, 30].

Despite their theoretical appeal, MOFS methods encounter severe operational limitations when scaled to ultra-high-dimensional biomedical data ($D > 10^3$):

1. **Combinatorial Archive and Sorting Overhead**: Maintaining an external non-dominated archive and executing Pareto ranking over dense high-dimensional vectors incurs substantial computational overhead, often exceeding 110–140 seconds in wall-clock time (as demonstrated in Section 5.7).
2. **Pareto Front Dilution in Ultra-High Dimensions**: In extreme $p \gg n$ regimes, non-dominated sorting tends to produce large, diffused Pareto fronts populated by non-sparse candidate subsets (often retaining 2,000–5,000 active features), failing to deliver the ultra-sparse compact biomarker signatures (< 100 features) required for clinical interpretability.

To circumvent Pareto archive bloat and achieve extreme subset sparsity within tight computational budgets, weighted single-objective scalarization remains the preferred formulation in high-dimensional biomarker discovery [12, 31]. However, conventional weighted frameworks enforce fixed static weights throughout search (e.g., $\mathbf{w} = [0.90, 0.10]$), which fails to adapt to the distinct evolutionary phases of feature selection. KDS-EL overcomes this limitation by introducing a dynamic Two-Phase Weight Scheduling strategy that aggressively eliminates noise features early and fine-tunes classification accuracy late.

2.3. Evolutionary Multi-Task Optimization in Feature Selection

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [32]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces. Prominent multi-task feature selection frameworks include minion-assisted multi-tasking [33], evolutionary multi-task PSO for hyperspectral and gene expression data [34, 35], multi-task bee colony band selection [36], and large-scale multi-objective multi-tasking [37].

Most notably, Zhang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, Subpopulation 1 performs continuous swarm exploration guided by inter-task attractors, while Subpopulation 2 conducts discrete feature sampling based on a shared feature-importance matrix. While MEL successfully demonstrates the power of cross-task knowledge transfer, its baseline implementation suffers from three critical bottlenecks: (i) random uniform initialization that causes a

Table 1: Structural and methodological comparison between KDS-EL and representative evolutionary feature selection algorithms.

Algorithm	Search Paradigm	Initialization Scheme	Knowledge Harvesting	Elite Refinement Mechanism	Objective Weighting	Evaluation Complexity
BPSO [7]	Single-Objective SI	Uniform Random	None (Memoryless)	Standard Swarm Update	Static Scalar	High (Unpruned)
FWPSO [8]	Single-Objective SI	Uniform Random	Static Filter Weights	Weight-Biased Velocity	Static Scalar	Moderate
SaWDE [9]	Single-Objective DE	Uniform Random	None	Self-Adaptive Mutation	Adaptive Scalar	High
W-MOPSO-FS [25]	Multi-Objective SI	Uniform Random	Filter-Wrapper Hybrid	Pareto Non-Dominated Archive	Pareto Front	High (Archive Overhead)
NNS-DFPG-CSA [40]	Single-Objective CSA	Hard MI Truncation	None	Dynamic Fractional-Order Guidance	Static Scalar	Moderate
MI-MPODE [39]	Single-Objective DE	MI Subspace Seeding	Static MI Ranking	Multi-Population Migration	Static Scalar	Moderate
EMSWOA [31]	Multi-Swarm WOA	Uniform Random	Dynamic Sub-Swarm Interaction	Elite Tuning Operator	Static Scalar	High
MEL [12]	Multi-Task SI	Uniform Random	Online Accuracy Feedback	Shared Importance Sampling	Static Scalar ($\mathbf{w} = [0.9, 0.1]$)	High (Sluggish Pruning)
KDS-EL (Proposed)	**Multi-Task SI**	**Soft MI Hybrid Seeding**	**Discounted Matrix W ($\gamma = 0.99$)**	**Knowledge-Guided KER (1-to-1)**	**Dynamic Two-Phase (TPS)**	**Low (3.4× Speedup)**

blind cold start in sparse spaces, (ii) unconstrained feature importance accumulation without memory discounting, and (iii) rigid static objective weights that delay early dimensionality reduction.

2.4. Knowledge-Guided Meta-Heuristics and Information-Theoretic Hybridization

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as a means to escape the blindness of black-box search [38–40]. In high-dimensional feature selection, mutual information (MI) serves as the primary information-theoretic metric to quantify non-linear feature relevance [41].

Existing knowledge-guided evolutionary algorithms typically adopt **one-shot offline injection**:

- **Static Filter Pre-Filtering**: Algorithms such as MI-MPODE [39] and NNS-DFPG-CSA [40] compute MI rankings offline and initialize the population or restrict the search space to the top-k ranked features.
- **Static Velocity Biasing**: Frameworks such as FWPSO [8] incorporate static feature-weight vectors to bias particle velocity equations throughout evolution.

While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates (such as neural networks [10] or Gaussian processes [42]) introduce severe surrogate training overhead and risk model misspecification in extreme $p \gg n$ regimes.

2.5. Unique Positioning and Structural Advancements of KDS-EL

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a **closed-loop, knowledge-guided evolutionary learning framework** that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling:

1. **Soft Filter-Wrapper Seeding vs. Hard Truncation**: Unlike conventional filter wrappers that permanently eliminate features via hard truncation, KDS-EL uses normalized MI scores as continuous Gaussian-perturbed seeds, preserving the full search space while ensuring an informative, sparse starting basin ($\|\mathbf{x}\|_0 \ll D/2$).
2. **Online Cumulative Knowledge Distillation vs. Memoryless Perturbation**: In contrast to black-box swarm mutations, KDS-EL dynamically harvests evolutionary fitness feedback into a cumulative importance matrix W, regulated by an exponential forgetting factor ($\gamma = 0.99$). The KER operator re-deploys this accumulated knowledge to execute targeted, directional elite flips within a strictly bounded computational budget ($k_e = 3 \ll N$).
3. **Dynamic Stage-Aware Objective Scheduling vs. Static Weights**: Unlike rigid static weighting ($\mathbf{w} = [0.90, 0.10]$), KDS-EL's Two-Phase Weight Scheduling temporally decouples coarse feature pruning (Phase 1, $w_f = 0.30$) from fine accuracy refinement (Phase 2, $w_a = 0.95$), achieving simultaneous gains in accuracy, feature compactness, and runtime efficiency.

Table 1 summarizes the structural comparison between KDS-EL and representative state-of-the-art evolutionary feature selection frameworks.

3. Preliminaries

This section establishes the theoretical foundations, mathematical notations, and baseline architectures that underpin our proposed framework. We first formulate classical continuous and binary particle swarm optimization, followed by the information-theoretic definition of mutual information for feature relevance estimation, and conclude with the dual-subpopulation multi-task evolutionary learning (MEL) baseline.

3.1. Particle Swarm Optimization and Binary Formulations

Particle Swarm Optimization (PSO), originally introduced by Kennedy and Eberhart [5], is a population-based stochastic optimization paradigm inspired by the collective foraging behaviors of bird flocks. A swarm consists of N particles navigating a D-dimensional search space $\mathbb{R}^D$. Each particle $i \in \{1, \ldots, N\}$ maintains a continuous position vector $\mathbf{p}_i^{(t)} = [p_{i,1}^{(t)}, \ldots, p_{i,D}^{(t)}]$ and a continuous velocity vector $\mathbf{v}_i^{(t)} = [v_{i,1}^{(t)}, \ldots, v_{i,D}^{(t)}]$ at iteration t. The historical trajectory is governed by its personal best position $\mathbf{p}_{\text{best},i}$ and the swarm's global best position $\mathbf{g}_{\text{best}}$.

The velocity and position updates are formulated as:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{1}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{2}$$

where ω denotes the inertia weight governing momentum retention [6], c_1 and c_2 are cognitive and social acceleration coefficients, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random stochastic factors.

To adapt PSO to combinatorial binary feature selection where solutions represent binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$ ($x_{i,j} = 1$ indicates feature j is selected), Kennedy and Eberhart [7] proposed the standard Binary PSO (BPSO). BPSO transforms continuous velocities into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \quad x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if rand()} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

While effective for low-dimensional problems, standard BPSO suffers severe exploration-exploitation dilemmas in ultra-high-dimensional spaces with $p \gg n$: continuous velocity updates provide no domain-grounded guidance toward sparse, informative biomarker subspaces. In KDS-EL, particles maintain continuous normalized position coordinates $\mathbf{p}_i \in [0, 1]^D$, mapped deterministically to binary feature masks via a strict thresholded Heaviside step function $x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$ with $\theta = 0.60$, eliminating stochastic binarization variance.

3.2. Mutual Information for Information-Theoretic Feature Relevance

Mutual information (MI), rooted in Shannon's information theory [41], quantifies the mutual dependence and reduction in uncertainty between two random variables. For a feature variable f_j and the discrete class label $Y \in \{1, \ldots, C\}$, the mutual information $I(f_j; Y)$ is defined as:

$$I(f_j; Y) = H(Y) - H(Y \mid f_j) = \sum_{v \in \text{Val}(f_j)} \sum_{c=1}^{C} P(f_j = v, Y = c) \log_2 \frac{P(f_j = v, Y = c)}{P(f_j = v) P(Y = c)}, \tag{4}$$

where $H(Y)$ is the marginal Shannon entropy of the class labels and $H(Y \mid f_j)$ is the conditional entropy given feature f_j. High values of $I(f_j; Y)$ indicate strong discriminative relevance to classification targets.

Algorithm 1 Traditional Filter Feature Selection via Hard MI Truncation

Require: Dataset $\mathcal{D} = (X, Y)$ with D features, top retention ratio $k \in (0, 1]$
Ensure: Truncated feature subset $\mathcal{S}_{\text{Filter}}$
1: **for** $j = 1$ **to** D **do**
2: Compute mutual information $I(f_j; Y)$ according to (4)
3: **end for**
4: Sort feature indices IDX in descending order of $I(f_j; Y)$
5: $\mathcal{S}_{\text{Filter}} \leftarrow \{\text{IDX}(1), \text{IDX}(2), \ldots, \text{IDX}(\lfloor k \cdot D \rfloor)\}$
6: **return** $\mathcal{S}_{\text{Filter}}$

Conventional filter feature selection methods [13, 40] evaluate $I(f_j; Y)$ across all D dimensions and execute a **hard truncation** (Algorithm 1),

discarding all features outside the top-k ranking. While computationally rapid ($O(D \log D)$), hard truncation fundamentally ignores multi-feature synergy: individual features with moderate MI scores that are collectively decisive when combined are permanently eliminated. Rather than using hard truncation, KDS-EL utilizes normalized MI scores as a **soft prior** to construct continuous initial population seeds (Section 4.2), injecting filter knowledge into the evolutionary wrapper without irreversibly shrinking the search space.

3.3. Multi-Task Evolutionary Learning Baseline (MEL)

Multi-Task Evolutionary Learning (MEL), recently proposed by Zhang et al. [12], introduces a cooperative multi-task architecture for high-dimensional feature selection. In MEL, a population of N particles is partitioned into two collaborative subpopulations, $\mathcal{P}_1$ and $\mathcal{P}_2$ ($|\mathcal{P}_1| = |\mathcal{P}_2| = N/2$), operating on dual complementary search tasks:

- **Subpopulation 1 ($\mathcal{P}_1$ - Continuous Swarm Exploration)**: Particles update continuous velocities and positions using PSO dynamics, augmented with an inter-task knowledge transfer channel directing particles toward the historical best of Subpopulation 2 ($\mathbf{g}_{\text{best},2}$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}). \tag{5}$$

- **Subpopulation 2 ($\mathcal{P}_2$ - Importance-Guided Discrete Sampling)**: Rather than continuous mechanics, $\mathcal{P}_2$ samples binary feature masks directly based on a shared feature-importance matrix $W_{\text{MEL}} \in \mathbb{R}^D$.

In MEL, both subpopulations share a scalar fitness function combining classification error and feature subset ratio via static weights:

$$f_{\text{MEL}}(\mathbf{x}) = \alpha \cdot (1 - \text{Acc}(\mathbf{x})) + \beta \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{6}$$

where $\text{Acc}(\mathbf{x})$ is the 5-fold cross-validation accuracy of a k-NN classifier ($k = 3$), and $\alpha = 0.90, \beta = 0.10$ are fixed constants [12].

Despite its innovative multi-task design, MEL exhibits three critical performance bottlenecks when scaled to ultra-high-dimensional data ($D > 10^3$):

1. **Blind Cold Start**: Uniform random initialization forces both subpopulations to start from dense configurations ($\approx 50\%$ active features), wasting substantial computational budget in non-informative spaces.
2. **Evolutionary Memory Loss and Stale Accumulation**: MEL's importance matrix is updated solely from per-generation accuracy differences without exponential decay, causing monotonic score accumulation that saturates ranking sensitivity.
3. **Static Objective Weight Rigidity**: The fixed weight $\beta = 0.10$ provides inadequate feature-pruning pressure during early generations, leading to sluggish feature reduction and prolonged evaluations (175.0 s average runtime).

KDS-EL directly inherits the multi-task dual-subpopulation structure of MEL while completely eliminating these three structural limitations through MI-guided hybrid seeding (Section 4.2), discounted cumulative importance harvesting with the KER refinement operator (Sections 4.3–4.4), and Two-Phase dynamic weight scheduling (Section 4.5).

4. Proposed KDS-EL Algorithm

In this section, we present the comprehensive architectural design, mathematical formulation, and algorithmic execution of the proposed **Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL)** algorithm. KDS-EL is designed specifically to overcome three fundamental structural deficiencies of conventional particle swarm optimization in ultra-high-dimensional feature selection: (i) the search inefficiency and wasted evaluation budget caused by uniform random initialization in sparse feature spaces, (ii) the loss of historical cross-iteration feature-importance signals due to memoryless random perturbation, and (iii) the inability of static objective weighting to adapt to the divergent requirements of coarse feature reduction versus fine-grained accuracy refinement.

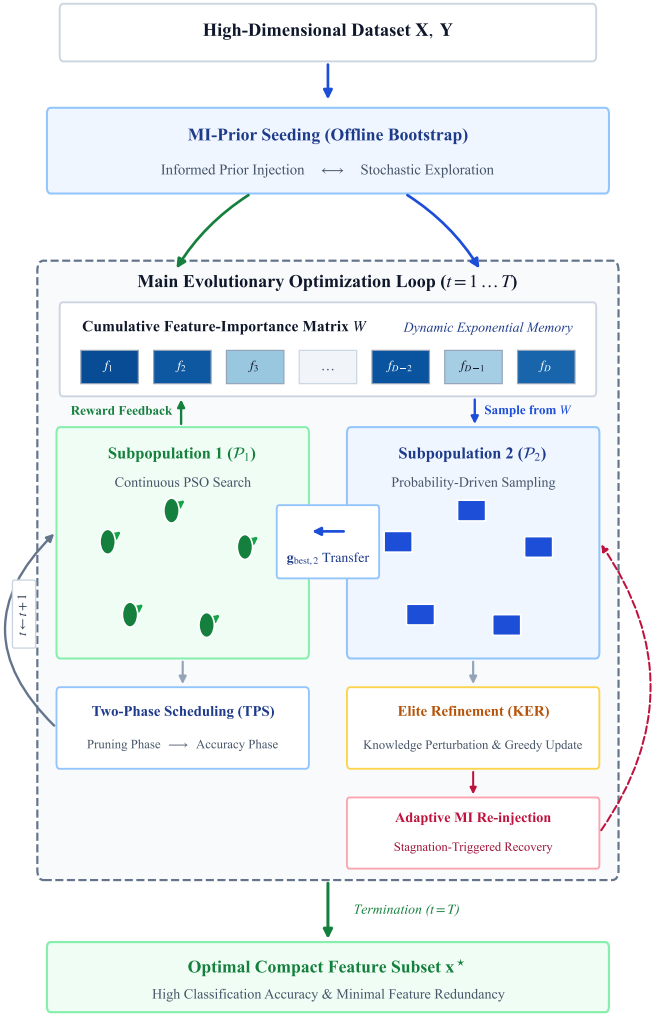


Figure 1: Overall architectural pipeline of KDS-EL: high-dimensional data undergo filter-based mutual information ranking, dual-subpopulation initialization with Gaussian perturbation, two-phase dynamic objective weight scheduling, cross-iteration cumulative feature-importance matrix accumulation with exponential decay, and knowledge-guided elite refinement with 1-to-1 in-place replacement.

4.1. Overall Algorithmic Architecture

The macroscopic architecture of KDS-EL is illustrated in Fig. 1. Built upon a cooperative multi-task dual-subpopulation backbone [12], KDS-EL divides a population of $N = 20$ binary particles into two collaborative subpopulations of equal size ($N_1 = N_2 = 10$). Subpopulation 1 ($\mathcal{P}_1$) executes standard continuous-space PSO velocity-position updates with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2. Subpopulation 2 ($\mathcal{P}_2$) performs probability-driven feature sampling guided by the population's accumulated feature-importance landscape.

The optimization process operates in a closed loop through four tightly coupled functional stages:

1. **Filter-Level Quality Bootstrapping (Offline)**: Mutual information (MI) between each feature and class labels is computed once prior to evolution, establishing a normalized baseline relevance vector $\mathbf{m} \in [0, 1]^D$.
2. **Knowledge-Informed Hybrid Seeding**: An initial population of N particles is generated by blending Gaussian-perturbed MI seeds with uniform random particles at a ratio of MI_init_ratio $= 0.50$, followed by Heaviside step binarization with threshold $\theta = 0.60$.
3. **Dynamic Two-Phase Evolutionary Loop**: Across $T = 100$ iterations, the objective weights $\mathbf{w}(t) = [w_a(t), w_f(t)]$ dynamically transition from aggressive feature pruning ($\mathbf{w}_1 = [0.70, 0.30]$ for $t \leq 0.7T$) to accuracy-oriented refinement ($\mathbf{w}_2 = [0.95, 0.05]$ for $t > 0.7T$).
4. **Accumulative Knowledge Harvesting and Elite Refinement**: Fitness improvements during evaluation are harvested into a continuous

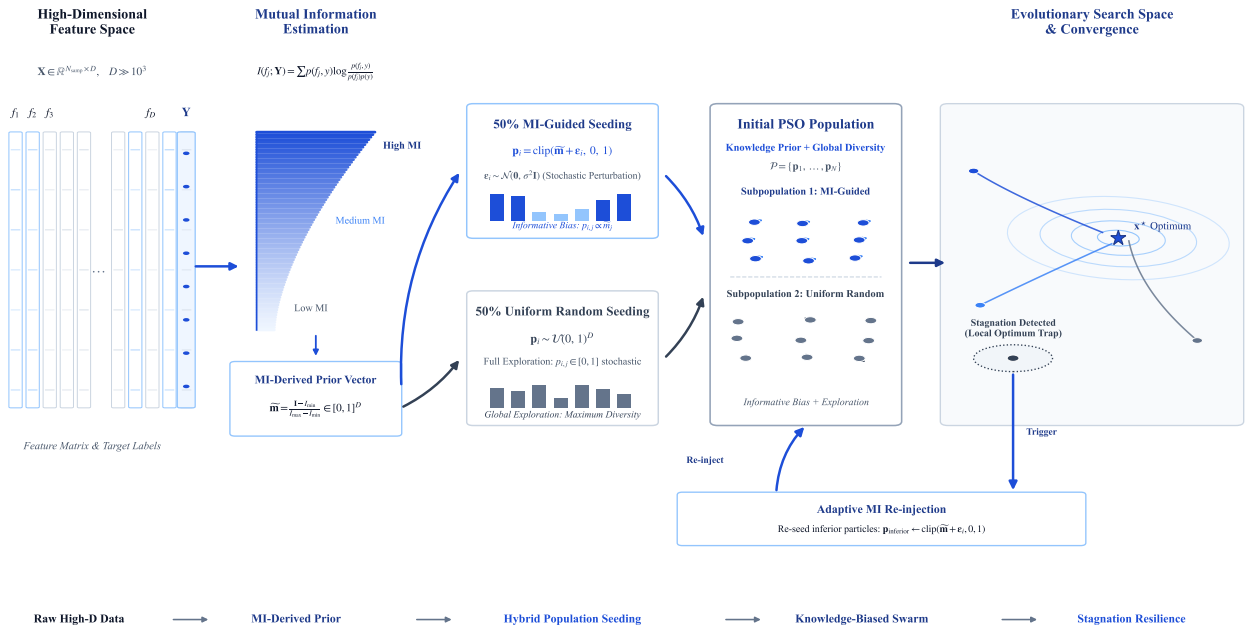


Figure 2: Operational mechanism of MI-Prior Seeding and Adaptive Re-injection: mutual information $I(f_j;\mathbf{Y})$ extracts continuous feature–target dependency to construct a normalized prior vector $\widetilde{\mathbf{m}} \in [0,1]^D$, which biases 50% of initial particles with Gaussian perturbation while the remaining 50% undergo uniform random exploration; when evolutionary search encounters stagnation (stall ≥ 20), the MI prior is adaptively re-injected into inferior particles to restore search vitality.

feature-importance matrix $W \in \mathbb{R}^D$, discounted by an exponential forgetting factor $\gamma = 0.99$. The top elite_k = 3 elite individuals are refined via the Knowledge-guided Elite Refinement (KER) operator and admitted via 1-to-1 in-place replacement. An adaptive re-initialization mechanism monitors stagnation and re-seeds the lower half of the population if global-best fitness stalls for stall_patience = 20 generations.

4.2. MI-Prior Seeding and Adaptive Re-injection Strategy

4.2.1. Deficiency Analysis and Design Rationale

Standard binary meta-heuristics universally initialize particle positions via uniform random sampling over $[0,1]^D$. In an ultra-high-dimensional feature space where $D \gg 10^3$, uniform random initialization produces candidate solutions where approximately 50% of all features are active ($\|\mathbf{x}\|_0 \approx D/2$). However, for biomedical classification problems, true informative biomarker subsets are notoriously sparse, typically comprising fewer than 100 genes ($\|\mathbf{x}^\star\|_0 \ll 0.01D$). Consequently, random initialization forces the swarm into dense, highly redundant, and noisy search basins, squandering a major fraction of the computational evaluation budget simply attempting to discard thousands of irrelevant features.

To resolve this initial cold-start inefficiency, KDS-EL establishes a filter-wrapper hybrid seeding strategy that injects prior feature-relevance knowledge into the swarm origin while preserving sufficient stochastic diversity.

4.2.2. Mathematical Formulation of MI Seeding

Let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote the training set with D-dimensional feature vectors $\mathbf{f}_i$ and class labels $y_i \in \{1,\ldots,C\}$. The mutual information $I(f_j;Y)$ between the j-th feature f_j and the label Y is defined as:

$$I(f_j;Y) = \sum_{v\in \text{Val}(f_j)} \sum_{c=1}^{C} P(f_j = v, Y = c) \log_2 \frac{P(f_j = v, Y = c)}{P(f_j = v)P(Y = c)}. \tag{7}$$

The continuous feature values are discretized into equal-frequency bins to estimate probability distributions. The raw MI scores $\{I(f_j;Y)\}_{j=1}^D$ are normalized into a baseline relevance vector $\mathbf{m} \in [0,1]^D$:

$$m_j = \frac{I(f_j;Y) - \min_{1\leq k\leq D} I(f_k;Y)}{\max_{1\leq k\leq D} I(f_k;Y) - \min_{1\leq k\leq D} I(f_k;Y) + \epsilon}, \tag{8}$$

where $\epsilon = 10^{-12}$ prevents division by zero.

To prevent population homogenization while guiding search toward informative dimensions, the initial population is split into two halves based on $\rho = \text{MI_init_ratio} = 0.50$:

$$\mathbf{p}_i = \begin{cases} \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), & \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_D), \quad \text{for } 1 \leq i \leq \rho N, \\ \mathbf{u}_i, & \mathbf{u}_i \sim \mathcal{U}(0,1)^D, \quad \text{for } \rho N < i \leq N, \end{cases} \tag{9}$$

where $\sigma = 0.10$ represents the Gaussian noise perturbation scale and $\text{clip}(z,0,1) = \max(0, \min(1,z))$.

Continuous positions $\mathbf{p}_i = [p_{i,1},\ldots,p_{i,D}]$ are mapped into binary feature selection vectors $\mathbf{x}_i \in \{0,1\}^D$ using a strict Heaviside step binarization with threshold $\theta = 0.60$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \geq \theta, \\ 0, & \text{if } p_{i,j} < \theta. \end{cases} \tag{10}$$

Because $\theta = 0.60 > 0.50$, only features possessing above-average mutual information are initialized as active, naturally yielding a sparse initial configuration ($\|\mathbf{x}_i\|_0 \approx 0.15D \sim 0.30D$) that accelerates early convergence. Algorithm 2 specifies the hybrid initialization procedure.

Algorithm 2 MI-Guided Population Initialization

Require: Normalized MI score vector $\mathbf{m} \in [0,1]^D$, population size $N = 20$, seeding ratio $\rho = 0.50$, threshold $\theta = 0.60$, noise scale $\sigma = 0.10$
Ensure: Hybrid binary initial population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^N$ and continuous positions $\{\mathbf{p}_i\}_{i=1}^N$

for $i = 1$ **to** N **do**
  if $i \leq \lfloor \rho N \rfloor$ **then**
    Sample perturbation $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_D)$
    $\mathbf{p}_i \leftarrow \max(\mathbf{0}, \min(\mathbf{1}, \mathbf{m} + \boldsymbol{\varepsilon}_i))$ ▷ Knowledge-guided continuous seed
  else
    Sample continuous position $\mathbf{p}_i \sim \mathcal{U}(0,1)^D$ ▷ Random exploratory seed
  end if
  for $j = 1$ **to** D **do**
    $x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta)$ ▷ Thresholded binarization (10)
  end for
end for
return Binary population $\{\mathbf{x}_i\}_{i=1}^N$ and continuous positions $\{\mathbf{p}_i\}_{i=1}^N$

4.2.3. Adaptive Stagnation Re-initialization

High-dimensional multimodal search landscapes frequently cause swarm particles to become trapped in suboptimal local attractors. When the global-best fitness f_{gbest} fails to improve for $\tau = \text{stall_patience} = 20$ consecutive iterations, an adaptive re-initialization routine is triggered:

1. The population is sorted in ascending order of personal-best fitness f_i^{pbest}.
2. The superior half ($\{i \mid \text{rank}(i) \le N/2\}$) is preserved to safeguard historical elite trajectories.
3. The inferior half ($\{i \mid \text{rank}(i) > N/2\}$) is fully refreshed using the Gaussian-perturbed MI distribution:

$$\mathbf{p}_i \leftarrow \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \ge \theta). \tag{11}$$

4. The refreshed particles are re-evaluated under the current objective weights, and the stagnation counter is reset to zero.

This restores diversity within a high-potential subspace without completely discarding established elite knowledge, as formalized in Algorithm 3.

Algorithm 3 Adaptive Stagnation Re-initialization

Require: Population $\{\mathbf{x}_i, \mathbf{p}_i\}_{i=1}^N$, personal bests $\{\mathbf{p}_{\text{best},i}, f_i^{\text{pbest}}\}_{i=1}^N$, MI vector $\mathbf{m}$, threshold $\theta = 0.60$, noise scale $\sigma = 0.10$, objective weights $\mathbf{w}(t)$
Ensure: Refreshed population $\mathcal{P}$ with restored swarm diversity

Sort population indices π such that $f_{\pi(1)}^{\text{pbest}} \le f_{\pi(2)}^{\text{pbest}} \le \cdots \le f_{\pi(N)}^{\text{pbest}}$
for $k = \lfloor N/2 \rfloor + 1$ **to** N **do**
  $i \leftarrow \pi(k)$ ▷ Target inferior particle
  Sample perturbation $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_D)$
  $\mathbf{p}_i \leftarrow \max(\mathbf{0}, \min(\mathbf{1}, \mathbf{m} + \boldsymbol{\varepsilon}_i))$
  $x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \ge \theta), \quad \forall j \in \{1, \ldots, D\}$
  Evaluate fitness $f_i \leftarrow f(\mathbf{x}_i, \mathbf{w}(t))$ using (22)
  $\mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i, \quad f_i^{\text{pbest}} \leftarrow f_i$
end for
return Updated population and personal bests

4.3. Cumulative Feature-Importance Matrix with Exponential Memory Decay

4.3.1. Deficiency Analysis and Design Rationale

Standard PSO operators update particle trajectories purely based on instantaneous personal best ($\mathbf{p}_{\text{best}}$) and global best ($\mathbf{g}_{\text{best}}$) coordinates. Across consecutive evolutionary cycles, thousands of candidate subset evaluations are performed, generating abundant empirical evidence regarding which specific feature additions or deletions consistently enhance classification performance. Standard meta-heuristics completely discard this trial-and-error feedback upon the conclusion of each generation, suffering from severe evolutionary memory loss.

To capture and retain this cross-iteration evolutionary feedback, KDS-EL maintains a global continuous **Cumulative Feature-Importance Matrix** $W \in \mathbb{R}^D$, initialized to the zero vector $\mathbf{0}$.

4.3.2. Differential Feedback Update and Exponential Forgetting

At generation t, whenever a newly evaluated particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its historical personal best ($f_i^{(t)} < f_i^{\text{pbest}}$), we extract the differential feature subset $C_i^{(t)}$ containing features whose selection status changed:

$$C_i^{(t)} = \{j \in \{1, \ldots, D\} \mid x_{i,j}^{(t)} \ne p_{\text{best},i,j}\}. \tag{12}$$

The magnitude of fitness improvement is quantified as:

$$\delta_i^{(t)} = f_i^{\text{pbest}} - f_i^{(t)} > 0. \tag{13}$$

Each changed feature $j \in C_i^{(t)}$ receives a directional reward proportional to $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{14}$$

where $(2x_{i,j}^{(t)} - 1) = +1$ if feature j was newly activated (rewarding beneficial inclusion) and -1 if feature j was deactivated (rewarding beneficial elimination). Features whose status remained unchanged, as well as non-improving particles ($\delta_i^{(t)} \le 0$), exert zero impact on W.

To prevent early-discovered features from dominating the matrix indefinitely and to allow the algorithm to forget stale historical correlations as the search shifts toward fine-grained subspaces, an exponential memory decay is applied to the entire matrix at the conclusion of each generation:

$$W \leftarrow \gamma \cdot W, \quad \text{where } \gamma = 0.99. \tag{15}$$

4.3.3. Theoretical Function of the Forgetting Factor γ

The exponential decay factor $\gamma = 0.99$ fulfills two indispensable theoretical roles:

- **Dynamic Rank Adaptation**: As the search progresses from coarse feature elimination to localized accuracy refinement, features that were critical in early generations may become redundant. The decay factor ensures that unless a feature repeatedly contributes to fitness improvements in recent generations, its cumulative score exponentially depreciates as γ^k, allowing newly discovered interactive features to rise in rank.

- **Score Saturation Prevention**: Without decay ($\gamma = 1.00$), cumulative importance values would grow monotonically unbounded ($W_j \to \infty$), causing the relative score distribution to saturate and destroying the sensitivity of downstream elite refinement operators.

Algorithm 4 outlines the per-generation importance harvesting and discounting procedure.

Algorithm 4 Feature-Importance Matrix Update and Decay

Require: Population candidates $\{\mathbf{x}_i\}_{i=1}^N$, personal bests $\{\mathbf{p}_{\text{best},i}, f_i^{\text{pbest}}\}_{i=1}^N$, candidate fitness $\{f_i\}_{i=1}^N$, current importance matrix $W \in \mathbb{R}^D$, decay factor $\gamma = 0.99$
Ensure: Discounted cumulative feature-importance matrix W

for $i = 1$ **to** N **do**
  if $f_i < f_i^{\text{pbest}}$ **then**
    Compute improvement magnitude $\delta_i \leftarrow f_i^{\text{pbest}} - f_i$
    for $j = 1$ **to** D **do**
      if $x_{i,j} \ne p_{\text{best},i,j}$ **then**
        $W_j \leftarrow W_j + \delta_i \cdot (2x_{i,j} - 1)$ ▷ Directional reward (14)
      end if
    end for
  end if
end for
$W \leftarrow \gamma \cdot W$ ▷ Exponential memory decay (15)
return Updated importance matrix W

4.4. Knowledge-guided Elite Refinement (KER) Strategy

4.4.1. Deficiency Analysis and Design Rationale

In conventional swarm algorithms, elite individuals (such as $\mathbf{g}_{\text{best}}$) are typically updated solely through global swarm convergence or perturbed via isotropic random bit mutations. In ultra-high-dimensional spaces with thousands of features, uniform random bit-flipping possesses an infinitesimal probability of flipping the precise subset of features required to escape a local optimum, resulting in stagnation during late evolutionary stages.

To exploit the historical evolutionary knowledge distilled in W, we introduce the **Knowledge-guided Elite Refinement (KER)** strategy. KER performs targeted, directional bit perturbations on top-performing solutions, selectively activating promising absent features and pruning uninformative active features.

4.4.2. Probabilistic Dual-Branch Formulation

At each generation, the top elite_k = 3 individuals (ranked by current fitness) are selected for refinement. For each elite individual $\mathbf{x}_e \in \{0, 1\}^D$, KER executes a probabilistic dual-branch operation (Fig. 3):

1. **Ergodic Random Exploration Branch (Probability $p_{\text{rand}} = 0.10$)**: To guarantee search ergodicity and prevent irreversible bias toward early-accumulated matrix patterns, $\lfloor \phi D \rfloor$ features are selected uniformly at random and inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = \lfloor \phi D \rfloor, \tag{16}$$

   where $\phi = \text{flip_frac} = 0.01$ defines the refinement perturbation budget.
2. **Knowledge-Guided Directional Refinement Branch (Probability $1 - p_{\text{rand}} = 0.90$)**: Each feature $j \in \{1, \ldots, D\}$ is assigned a state-dependent directional flip score σ_j:

$$\sigma_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0 \quad \text{(reward activating absent high-importance features)}, \\ -W_j, & \text{if } x_{e,j} = 1 \quad \text{(reward deactivating present low-importance features)}. \end{cases} \tag{17}$$

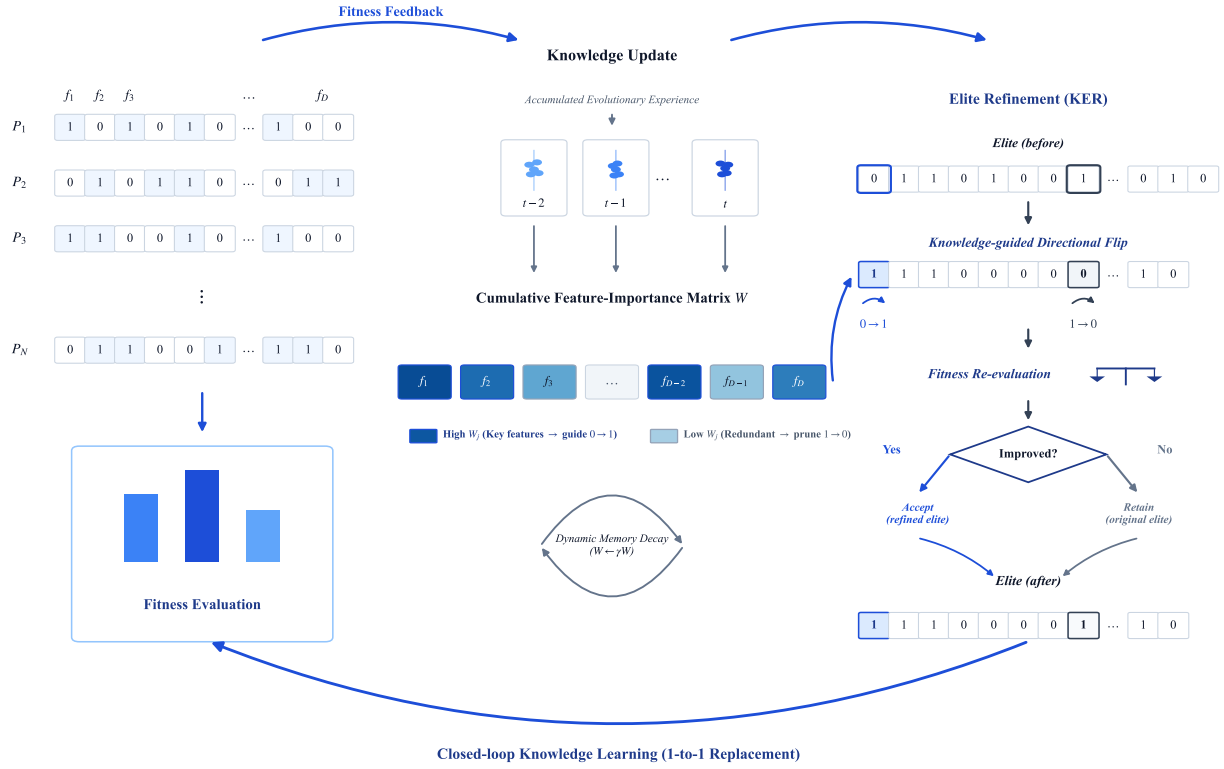


Figure 3: Operational workflow of the Knowledge-guided Elite Refinement (KER) strategy: candidate elites undergo either knowledge-guided directional refinement ($p = 0.90$) guided by cumulative matrix W or ergodic random exploration ($p = 0.10$), evaluated in a closed-loop and admitted via greedy 1-to-1 in-place replacement.

The scores can be written compactly as $\sigma_j = (1-2x_{e,j})\cdot W_j$. We identify the index set $\mathcal{S}_{\text{know}}$ containing the $\lfloor \phi D \rfloor$ features possessing the highest flip scores σ_j:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S}\subset\{1,\dots,D\} \\ |\mathcal{S}|=\lfloor \phi D \rfloor}} \sum_{j\in\mathcal{S}} \sigma_j. \tag{18}$$

The directional refinement is then applied:

$$x'_{e,j} = \begin{cases} 1 - x_{e,j}, & \text{if } j \in \mathcal{S}_{\text{know}}, \\ x_{e,j}, & \text{otherwise.} \end{cases} \tag{19}$$

4.4.3. 1-to-1 In-Place Replacement and Bounded Evaluation Budget

The perturbed offspring $\mathbf{x}'_e$ is evaluated under current objective weights $\mathbf{w}(t)$. A greedy **1-to-1 in-place replacement rule** is enforced:

$$\mathbf{x}^\star_e = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{20}$$

If improved, $\mathbf{x}'_e$ immediately overwrites $\mathbf{x}_e$ in the population and updates the global best if $f(\mathbf{x}'_e) < f_{\text{gbest}}$.

Rationale over $2N$-Candidate Pooling: Unlike early multi-task prototypes that maintain an expanded $2N$ competitive selection pool, 1-to-1 in-place replacement incurs zero archive management overhead and prevents high selection pressure from prematurely collapsing population diversity.

Bounded Refinement Budget: Because elite_k = 3 is strictly fixed and $\phi = 0.01$ modifies only 1% of the feature dimensions, KER executes within an exact bounded evaluation overhead of $k_e = 3$ function evaluations per generation, avoiding exponential computational expansion. Algorithm 5 details the KER operator.

Algorithm 5 Knowledge-guided Elite Refinement (KER)

Require: Elite candidate $\mathbf{x}_e$, importance matrix W, perturbation fraction $\phi = 0.01$, random branch probability $p_{\text{rand}} = 0.10$, objective weights $\mathbf{w}(t)$
Ensure: Refined elite solution $\mathbf{x}^\star_e$ satisfying $f(\mathbf{x}^\star_e) \le f(\mathbf{x}_e)$

```
Sample random variable r ~ U(0, 1)
k_flip ← max(1, ⌊φD⌋)
if r < p_rand then                          ▷ 10% Ergodic random exploration branch
    Sample subset S_rand ⊂ {1, ..., D} uniformly at random with |S_rand| = k_flip
    x'_e ← x_e;   x'_{e,j} ← 1 − x_{e,j}, ∀j ∈ S_rand
else                                        ▷ 90% Knowledge-guided directional refinement branch
    for j = 1 to D do
        σ_j ← (1 − 2x_{e,j}) · W_j          ▷ Compute directional score (17)
    end for
    Find subset S_know of size k_flip with top σ_j values (18)
    x'_e ← x_e;   x'_{e,j} ← 1 − x_{e,j}, ∀j ∈ S_know
end if
Evaluate offspring fitness f' ← f(x'_e, w(t)) using (22)
if f' < f(x_e, w(t)) then
    x*_e ← x'_e                             ▷ In-place elite replacement (20)
else
    x*_e ← x_e
end if
return x*_e
```

4.5. Two-Phase Dynamic Objective Weight Scheduling Strategy

4.5.1. Deficiency Analysis and Design Rationale

In multi-objective feature selection transformed via weighted-sum aggregation, conventional algorithms universally fix the objective weights throughout the entire evolutionary trajectory (e.g., static $\mathbf{w} = [0.90, 0.10]$) [12, 31]. However, feature selection exhibits an intrinsic temporal asymmetry between early and late search stages:

- **Early Evolutionary Stage**: The swarm navigates dense, highly noisy regions containing thousands of redundant features. The primary optimization objective is rapid, aggressive dimensionality reduction to prune non-informative search space.
- **Late Evolutionary Stage**: Once redundant features have been aggressively stripped and the swarm converges to a compact candidate set (< 100 features), the objective shifts decisively toward fine-grained accuracy tuning among subtle biomarker combinations.

A static weight vector compromises between these competing stage-dependent



Figure 4: Evolutionary optimization mechanism of the Two-Phase Dynamic Objective Weight Scheduling (TPS) strategy: across a continuous scalarized fitness landscape, search transitions at $t = 0.70\,T$ from Phase 1 ($(\alpha, \beta) = (0.70, 0.30)$, broad exploration and progressive feature reduction) to Phase 2 ($(\alpha, \beta) = (0.95, 0.05)$, focused accuracy refinement), ultimately converging to a high-quality compact feature subset.

goals, leading to sluggish early feature reduction and inadequate late-stage accuracy refinement.

4.5.2. Dynamic Weight Scheduling Formulation

To reconcile this temporal conflict, KDS-EL introduces the **Two-Phase Weight Scheduling (TPS)** strategy, as illustrated in Fig. 4. The total generational budget $T = 100$ is partitioned at split point $\alpha = \text{PhaseSplit} = 0.70$. The dynamic dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$ is governed by a step-transition function:

$$\mathbf{w}(t) = \begin{cases} [0.70,\ 0.30], & \text{for } 1 \le t \le \lfloor \alpha T \rfloor \quad \text{(Phase 1: Aggressive Feature Pruning)}, \\ [0.95,\ 0.05], & \text{for } \lfloor \alpha T \rfloor < t \le T \quad \text{(Phase 2: Accuracy Refinement)}. \end{cases} \tag{21}$$

The unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0,1\}^D$ at generation t is formulated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot \text{Error}(\mathbf{x}) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{22}$$

where $\text{Error}(\mathbf{x}) = 1 - \text{Acc}(\mathbf{x})$ denotes the 5-fold cross-validation classification error rate evaluated by a k-nearest neighbors (k-NN, $k = 3$) classifier, and $\|\mathbf{x}\|_0 / D = \frac{1}{D} \sum_{j=1}^{D} x_j$ represents the normalized feature selection ratio (FSR).

In Phase 1 ($t \le 70$), the heavy feature-ratio penalty ($w_f = 0.30$) drives rapid elimination of thousands of non-informative features without sacrificing classification fidelity. In Phase 2 ($t > 70$), the weight switches to prioritize classification accuracy ($w_a = 0.95$), allowing the swarm and the KER operator to fine-tune delicate biomarker interactions over the surviving compact subspace. Algorithm 6 specifies the TPS scheduling function.

Algorithm 6 Two-Phase Dynamic Weight Scheduling

Require: Current generation t, maximum generations $T = 100$, phase split ratio $\alpha = 0.70$
Ensure: Dynamic dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$

```
if t ≤ ⌊α · T⌋ then
    w(t) ← [0.70, 0.30]          ▷ Phase 1: Coarse feature pruning
else
    w(t) ← [0.95, 0.05]          ▷ Phase 2: Fine accuracy refinement
end if
return w(t)
```

4.6. Integrated KDS-EL Main Algorithm

Algorithm 7 consolidates the holistic execution workflow of KDS-EL, integrating filter-based MI ranking, hybrid seeding, the two-phase evolutionary loop, cumulative importance matrix updating, and knowledge-guided elite refinement into a unified procedure.

4.7. Asymptotic Computational and Space Complexity Analysis

To establish the computational efficiency of KDS-EL, we conduct a rigorous asymptotic complexity analysis covering all algorithmic components.

4.7.1. Time Complexity Decomposition

The execution time of KDS-EL is decomposed into two distinct phases:

1. **Offline Preprocessing Phase (Mutual Information Calculation)**: Computing the mutual information $I(f_j; Y)$ between each feature f_j and class label Y requires $O(N_{\text{samp}})$ operations per feature, where N_{samp} is the number of training samples. For all D features, computing and sorting the baseline MI vector incurs:

$$T_{\text{pre}} = O(D \cdot N_{\text{samp}} + D \log D). \tag{23}$$

   Because this step is executed only once prior to the evolutionary loop, its overhead is entirely negligible relative to iterative wrapper evaluations.

2. **Iterative Evolutionary Main Loop (T Iterations)**: In each of the T generations, computational costs arise from four operations:

   - *Subpopulation Position Updates*: Updating particle velocities and continuous positions across the population of size N incurs $O(N \cdot D)$ arithmetic operations.
   - *Fitness Wrapper Evaluation*: Evaluating the 5-fold cross-validation k-NN classification accuracy for N candidate particles requires:

$$T_{\text{eval}} = O(N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{24}$$

   where $d_{\text{selected}} = \|\mathbf{x}\|_0$ is the number of active features in candidate $\mathbf{x}$.

Algorithm 7 Complete Execution Workflow of KDS-EL

Require: Dataset $\mathcal{D} = (X, Y)$ with D features, population size $N = 20$, maximum iterations $T = 100$, split ratio $\alpha = 0.70$, binarization threshold $\theta = 0.60$, MI ratio $\rho = 0.50$, decay $\gamma = 0.99$, elite count $k_e = 3$, flip fraction $\phi = 0.01$, random branch probability $p_{\text{rand}} = 0.10$, stall patience $\tau = 20$

Ensure: Optimal binary feature subset $\mathbf{x}^\star \in \{0,1\}^D$

/* Stage 1: Filter-level Feature Ranking */
Compute mutual information $I(f_j; Y)$ for all $j \in \{1, \ldots, D\}$ via (7)
Normalize MI scores to construct relevance vector $\mathbf{m} \in [0,1]^D$ via (8)
/* Stage 2: Knowledge-Guided Population Seeding */
Initialize population $\{\mathbf{x}_i, \mathbf{p}_i\}_{i=1}^N$ via Algorithm 2 (MI-guided hybrid seeding)
Initialize cumulative feature-importance matrix $W \leftarrow \mathbf{0} \in \mathbb{R}^D$
Evaluate initial fitness $\{f_i\}_{i=1}^N$ under $\mathbf{w}(1) = [0.70, 0.30]$ via (22)
Initialize personal bests: $\mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i,\ f_i^{\text{pbest}} \leftarrow f_i,\ \forall i \in \{1, \ldots, N\}$
Identify global best: $\mathbf{g}_{\text{best}} \leftarrow \arg\min_{\mathbf{p}_{\text{best},i}} f_i^{\text{pbest}},\ f_{\text{gbest}} \leftarrow \min f_i^{\text{pbest}}$
Initialize stagnation counter stall_count $\leftarrow 0$
/* Stage 3: Two-Phase Evolutionary Main Loop */
for $t = 1$ **to** T **do**
  Update objective weights $\mathbf{w}(t) \leftarrow$ Algorithm 6(t, T, α)
  /* Subpopulation 1: Inter-Task Velocity-Position Update */
  for $i = 1$ **to** $N/2$ **do**
    Update continuous velocity $\mathbf{v}_i$ and position $\mathbf{p}_i$ using PSO rules with Subpopulation 2 attractor
    Binarize position: $x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta),\ \forall j \in \{1, \ldots, D\}$
  end for
  /* Subpopulation 2: Importance-Guided Feature Sampling */
  for $i = N/2 + 1$ **to** N **do**
    Sample binary features based on cumulative matrix W and personal best $\mathbf{p}_{\text{best},i}$
  end for
  /* Fitness Evaluation and Importance Matrix Harvesting */
  for $i = 1$ **to** N **do**
    Evaluate candidate fitness $f_i \leftarrow f(\mathbf{x}_i, \mathbf{w}(t))$ using (22)
  end for
  Update and discount importance matrix $W \leftarrow$ Algorithm 4$(\{\mathbf{x}_i\}, \{\mathbf{p}_{\text{best},i}\}, \{f_i\}, \{f_i^{\text{pbest}}\}, W, \gamma)$
  Update personal bests: if $f_i < f_i^{\text{pbest}}$ then $\mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i,\ f_i^{\text{pbest}} \leftarrow f_i$
  /* Stage 4: Knowledge-Guided Elite Refinement */
  Identify top k_e elite indices $\mathcal{E} \subset \{1, \ldots, N\}$ based on f_i^{pbest}
  for each $e \in \mathcal{E}$ **do**
    $\mathbf{p}_{\text{best},e} \leftarrow$ Algorithm 5$(\mathbf{p}_{\text{best},e}, W, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ 1-to-1 in-place refinement
  end for
  /* Global Best Update and Stagnation Check */
  Find current best $\mathbf{x}_{\text{curr}} \leftarrow \arg\min_{\mathbf{p}_{\text{best},i}} f_i^{\text{pbest}}$ with fitness f_{curr}
  if $f_{\text{curr}} < f_{\text{gbest}}$ **then**
    $\mathbf{g}_{\text{best}} \leftarrow \mathbf{x}_{\text{curr}},\ f_{\text{gbest}} \leftarrow f_{\text{curr}},$ stall_count $\leftarrow 0$
  else
    stall_count $\leftarrow$ stall_count + 1
  end if
  if stall_count $\geq \tau$ **then**
    Execute adaptive re-initialization via Algorithm 3
    stall_count $\leftarrow 0$
  end if
end for
return $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$

- *Importance Matrix Differential Updating and Decay*: Scanning changed features for improving particles and performing scalar decay across matrix W incurs $O(N \cdot D + D)$ operations.
- *Knowledge-Guided Elite Refinement (KER)*: Computing flip scores σ_j for k_e elites, selecting the top $\lfloor \phi D \rfloor$ features, flipping bits, and re-evaluating the k_e refined elites requires $O(k_e \cdot (D + T_{\text{eval_single}}))$, where $T_{\text{eval_single}} = O(N_{\text{samp}}^2 \cdot d_{\text{selected}})$.
- *Adaptive Stagnation Re-initialization*: When triggered (at most $\lfloor T/\tau \rfloor$ times), re-seeding and evaluating $N/2$ particles requires $O(\frac{N}{2} \cdot (D + T_{\text{eval_single}}))$.

Summing these terms across T generations yields the total asymptotic time complexity:

$$\mathcal{T}_{\text{total}} = O\Big(D \log D + T \cdot [N \cdot D + (N + k_e) \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}]\Big). \tag{25}$$

4.7.2. Theoretical Complexity Equivalence and Practical Acceleration

Because $k_e = 3 \ll N = 20$ and $N_{\text{samp}}^2 \cdot d_{\text{selected}} \gg D$, the dominant asymptotic per-generation complexity of KDS-EL is strictly governed by wrapper evaluations:

$$\mathcal{T}_{\text{KDS-EL}} = O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{26}$$

which is identically of the same asymptotic order as the baseline MEL [12] ($O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{MEL}})$).

Crucially, in the baseline MEL, random initialization and sluggish feature reduction result in dense candidate subsets ($d_{\text{MEL}} \approx 2{,}722$ active features on average). In contrast, KDS-EL's MI seeding and Phase 1 aggressive feature pruning rapidly compress the active feature count to $d_{\text{KDS-EL}} \approx 88$ features. Because k-NN distance calculation scales linearly with d_{selected}, evaluating a KDS-EL particle is over an order of magnitude faster than evaluating an unpruned MEL particle. Consequently, KDS-EL delivers a dramatic **3.4× empirical wall-clock speedup** (51.4 s vs. 175.0 s) while incorporating richer knowledge operators.

4.7.3. Space Complexity

The memory footprint of KDS-EL comprises: (i) the population continuous and binary position matrices ($O(N \cdot D)$), (ii) personal-best vectors ($O(N \cdot D)$), (iii) the cumulative feature-importance matrix W ($O(D)$), and (iv) the precomputed normalized MI vector $\mathbf{m}$ ($O(D)$). The overall space complexity is:

$$\mathcal{S}_{\text{total}} = O(N \cdot D + N_{\text{samp}} \cdot D), \tag{27}$$

which scales strictly linearly with dimension D, ensuring minimal memory consumption.

5. Experimental Results and Analysis

This section reports the experimental outcomes across six complementary perspectives: overall classification accuracy, feature-selection compactness, statistical significance, ceiling-effect behavior, convergence dynamics, and computational efficiency. All aggregate numbers are computed over the 20 benchmark datasets described in Section 5.1. We then quantify the contribution of each mechanism in Section 5.8 and assess the sensitivity of KDS-EL to its hyper-parameters in Section 5.9.

5.1. Experimental Setup

This subsection outlines the experimental framework designed to rigorously assess the performance, compactness, scalability, and robustness of KDS-EL. To provide a comprehensive evaluation, we establish a standardized testbed spanning benchmark datasets across diverse domains, state-of-the-art comparison algorithms, uniform experimental configurations, and a formal statistical validation protocol.

5.1.1. Benchmark Datasets and Domain Characteristics

The empirical evaluation is conducted on 20 widely recognized benchmark datasets drawn from five distinct real-world domains: 15 biomedical gene-expression microarrays (*ALL1*, *ALL4*, *ALL_AML*, *Adenoma*, *Colon*, *DLBCL*, *Gastric*, *Gastric1*, *Gastric2*, *Leukaemia*, *Lymphoma*, *Myeloma*, *Prostate*, *SR-BCT*, and *T1D*), two text categorization corpora (*BASEHOCK* and *PCMAC*), one computer vision image dataset (*COIL20*), one spoken speech dataset (*ISOLET*), and one synthetic classification benchmark (*madelon*). The comprehensive characteristics of these datasets are summarized in Table 3.

The benchmark suite embodies several critical challenges inherent to high-dimensional feature selection:

1. **Extreme dimensionality and the $p \gg n$ bottleneck**: The feature dimensionality D ranges from 500 (*madelon*) to 54,675 (*T1D*), while instance counts N_{samp} vary between 36 (*Adenoma*) and 2,600 (*madelon*). Crucially, 17 out of the 20 datasets exhibit the extreme $p \gg n$ regime where the dimensionality exceeds the sample size by several orders of magnitude, making conventional machine learning models severely prone to overfitting and the curse of dimensionality.

2. **Diverse class structures and data modalities**: The collection spans 17 binary classification problems as well as multi-class tasks, including *SRBCT* (4 classes), *COIL20* (20 classes), and *ISOLET* (26 classes), testing algorithmic generalization across continuous microarray expressions, sparse text frequencies, and dense pixel representations.
3. **Presence of accuracy ceiling datasets**: The suite explicitly includes eight gene-expression datasets—*ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*—on which competitive feature selection methods routinely achieve near-perfect classification accuracy exceeding 0.998. On these datasets, classification accuracy saturates, and the number of selected features serves as the primary discriminant of algorithmic effectiveness, as analyzed in Section 5.5.

To prevent data leakage during feature relevance estimation and evaluation, a stratified 5-fold cross-validation scheme with random shuffling is strictly enforced. The cross-validation partitioning is performed prior to feature selection, ensuring that candidate subsets are evaluated strictly on validation folds.

5.1.2. Comparison Algorithms and Baseline Categorization

To evaluate KDS-EL comprehensively against modern evolutionary feature selection paradigms, 14 fixed-length methods are compared under identical evaluation budgets. As categorized in Table 2, the competitor suite comprises three representative groups:

1. **Proposed framework and ablative configurations**: The complete KDS-EL and its systematically stripped variants:
   - *KDS-EL*: The full framework integrating MI-guided initialization, the KER strategy, and Two-Phase Weight Scheduling (TPS).
   - *KDS-EL w/o TPS*: The framework without two-phase scheduling, employing a static objective weighting $\mathbf{w}_s = [0.9, 0.1]$ with in-place elite replacement.
   - *KDS-EL w/o TPS&KER*: The framework stripped of both two-phase scheduling and the KER strategy, corresponding to the MI-initialized multi-task baseline (MEL+MI).
   - *MEL* [12]: The foundational multi-task evolutionary learning baseline stripped of all three mechanisms (random initialization, static weighting, no elite refinement).
2. **State-of-the-art single-objective evolutionary algorithms**: Five prominent evolutionary wrappers tailored for high-dimensional feature selection:
   - *FWPSO* [8]: Feature-weighting particle swarm optimization integrating feature saliency into velocity updates.
   - *PSO-CSM* [11]: A recently proposed PSO algorithm employing a comprehensive scoring framework.
   - *SaWDE* [9]: Self-adaptive weighted differential evolution for large-scale feature selection.
   - *IGPSO* [10]: Importance-guided PSO leveraging a multilayer perceptron for relevance estimation.
   - *BPSO* [7]: The classical binary particle swarm optimization algorithm.
3. **Multi-objective evolutionary feature selection (MOFS) algorithms**: Five multi-objective methods that simultaneously optimize classification accuracy and subset size:
   - *W-MOPSO-FS*: A self-built Wrapper–Filter hybrid multi-objective PSO.
   - *fMOPSO-FS* [43]: Self-adaptive multi-objective PSO based on feature-label correlation.
   - *NSGA-II-FS* [26]: The non-dominated sorting genetic algorithm applied to binary subset selection.
   - *MOEA-D-FS* [27]: Decomposition-based multi-objective evolutionary algorithm for feature selection.
   - *MOPSO-FS* [28]: Standard multi-objective PSO with external Pareto archive management.

All comparison algorithms are implemented based on their original publications, adopting the hyper-parameter settings recommended by their respective authors.

5.1.3. Parameter Configurations and Experimental Environment

To guarantee fairness across all comparative experiments, a uniform experimental protocol is applied. The K-nearest neighbor (KNN) classifier with $k = 3$ serves as the standard evaluation model across all algorithms due to its non-parametric nature and sensitivity to feature subset quality. Each algorithm is executed with a swarm population size of $N = 20$ candidate solutions over a maximum of $T = 100$ iterations. For continuous-to-binary transformation in continuous velocity-based algorithms, the standard threshold $\theta = 0.6$ is employed. Every experiment is repeated over 10 independent runs with distinct random seeds.

The specific hyper-parameter settings for KDS-EL are configured as follows: knowledge weight decay factor WeightDecay = 0.99; phase split ratio PhaseSplit = 0.70 with stage-wise objective weights $\mathbf{w}_s = [0.70, 0.30]$ during Phase 1 (aggressive feature reduction) and $\mathbf{w}_s = [0.95, 0.05]$ during Phase 2 (fine accuracy refinement); KER perturbation fraction flip_frac = 0.01; KER random exploration probability rand_flip_p = 0.10; elite refinement pool size elite_k = 3; MI initialization ratio MI_init_ratio = 0.50; and adaptive stagnation patience stall_patience = 20 generations. Table 4 provides the complete summary of parameter settings. All experiments were conducted on a workstation equipped with an Intel Core i5-13700KF CPU @ 3.40 GHz and 32 GB RAM under Windows 11 and MATLAB R2023b / Python 3.9 execution environments.

5.1.4. Performance Metrics and Statistical Significance Framework

Algorithm performance is evaluated across four quantitative dimensions:

1. **Mean Classification Accuracy** (Acc): The average 5-fold cross-validation accuracy over 10 independent runs, defined as
$$\mathrm{Acc} = \frac{1}{R}\sum_{r=1}^{R}\mathrm{Acc}_r, \qquad R = 10, \tag{28}$$
accompanied by its standard deviation (Std) to measure solution stability.
2. **Mean Selected Feature Count** ($\bar{d}$): The average number of active dimensions in the final feature mask:
$$\bar{d} = \frac{1}{R}\sum_{r=1}^{R}\|\mathbf{x}_r^*\|_0, \tag{29}$$
along with the Feature Selection Ratio (FSR = $\frac{\bar{d}}{D} \times 100\%$).
3. **Computational Runtime** ($\bar{t}$): The average wall-clock execution time in seconds per single run.
4. **Statistical Significance Testing Protocol**: To verify whether observed improvements are statistically meaningful, we adhere to the multi-dataset statistical testing protocol recommended by Demšar [44]:
   - *Friedman Mean Rank Test* [45]: Computed across the 20×14 accuracy matrix to determine global algorithm rankings (lower rank indicates superior performance), with tied per-dataset means assigned their average rank.
   - *Pairwise Wilcoxon Signed-Rank Test* [44]: Conducted at the $\alpha = 0.05$ significance level to assess pairwise superiority between KDS-EL and each of the 13 competing methods across the 20 benchmark datasets.
   - *Holm–Bonferroni Step-Down Correction* [46]: Applied to adjust the family-wise error rate across all 13 pairwise hypothesis tests, ensuring rigorous statistical validation against multiple-comparison artifacts.
   - *Per-dataset Paired Signed-Rank Annotation*: In Table 5, each competitor cell is annotated with $+$, $\approx$, or $-$ based on a two-sided Wilcoxon signed-rank test ($p < 0.05$) across the 10 paired runs, indicating whether KDS-EL is significantly superior, statistically equivalent, or significantly inferior on that individual dataset.

5.2. Overall Classification Accuracy and Comparative Analysis

Table 5 reports the mean classification accuracy along with standard deviations over 10 independent runs across the 20 benchmark datasets for all 14 compared methods. The experimental results demonstrate that KDS-EL achieves superior classification performance, securing the highest overall mean accuracy of **0.9574**.

Compared with the baseline and ablative variants, KDS-EL outperforms its fixed-weight counterpart without two-phase scheduling (KDS-EL w/o

Table 2: The 14 fixed-length methods compared in this study.

Method	Category	Reference
KDS-EL	Proposed	This work
KDS-EL w/o TPS	Ablation	This work
KDS-EL w/o TPS&KER	Ablation	This work
MEL	Baseline	[12]
FWPSO	Single-objective	[8]
PSO-CSM	Single-objective	[11]
SaWDE	Single-objective	[9]
IGPSO	Single-objective	[10]
BPSO	Single-objective	[7]
W-MOPSO-FS	Multi-objective	Self-built
fMOPSO-FS	Multi-objective	[43]
NSGA-II-FS	Multi-objective	[26]
MOEA-D-FS	Multi-objective	[27]
MOPSO-FS	Multi-objective	[28]

Table 3: Dataset Description (20 Benchmark Datasets)

Dataset	Samples	Features	Classes	Domain
Adenoma	36	7,457	2	Gene expression
ALL_AML	72	7,129	2	Gene expression
ALL1	128	12,625	2	Gene expression
ALL4	93	12,625	2	Gene expression
Colon	62	2,000	2	Gene expression
DLBCL	77	7,129	2	Gene expression
Gastric	65	22,645	2	Gene expression
Gastric1	144	22,283	2	Gene expression
Gastric2	124	22,283	2	Gene expression
Leukaemia	72	7,129	2	Gene expression
Lymphoma	45	4,026	2	Gene expression
Myeloma	173	12,625	2	Gene expression
Prostate	102	12,625	2	Gene expression
SRBCT	83	2,308	4	Gene expression
T1D	101	54,675	2	Gene expression
BASEHOCK	1,993	4,862	2	Text
COIL20	1,440	1,024	20	Image
ISOLET	1,560	617	26	Speech
madelon	2,600	500	2	Synthetic
PCMAC	1,943	3,289	2	Text

Range: $n \in [36, 2600]$, $d \in [500, 54675]$, $c \in [2, 26]$

Table 4: Parameter Settings

Common Experimental Setup		
Population size (N)	=	20
Maximum iterations (T)	=	100
KNN neighbors (k)	=	3
Real-to-binary threshold (θ)	=	0.6
Cross-validation	=	5-fold stratified, shuffled
Number of independent runs	=	10
KDS-EL (full framework)		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
Knowledge weight decay	=	0.99
Elite flip fraction	=	0.01
Elite count (k_e)	=	3
Random flip probability	=	0.10
MI-guided init ratio	=	0.5
Stall patience (P_s)	=	20
Phase split ratio	=	0.7
Phase-1 objective weights	=	$\mathbf{w}_s = [0.7, 0.3]$
Phase-2 objective weights	=	$\mathbf{w}_s = [0.95, 0.05]$
KDS-EL w/o TPS (no two-phase scheduling)		
Fixed objective weights	=	$\mathbf{w}_s = [0.9, 0.1]$
MEL and KDS-EL w/o TPS&KER		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
MI-guided init ratio (KDS-EL w/o TPS&KER)	=	0.5
Stall patience (KDS-EL w/o TPS&KER)	=	20
Competitor Methods		
Competitors (BPSO, FWPSO, IGPSO, PSO-CSM, SaWDE, NSGA-II-FS, MOPSO-FS, MOEA-D-FS, fMOPSO-FS, W-MOPSO-FS) use the hyper-parameters recommended in their original publications, with $N = 20$ and $T = 100$ unless the original paper specifies otherwise.		

TPS) by **+0.86 pp** (0.9574 vs. 0.9488), the MI-initialized variant without elite refinement (KDS-EL w/o TPS&KER) by **+2.49 pp** (0.9574 vs. 0.9325), and the original MEL baseline [12] by a substantial margin of **+7.06 pp** (0.9574 vs. 0.8868). Against established single-objective state-of-the-art competitors, KDS-EL surpasses the strongest single-objective method FWPSO [8] by **+3.97 pp** (0.9574 vs. 0.9177), PSO-CSM [11] by **+5.53 pp** (0.9574 vs. 0.9021), SaWDE [9] by **+6.30 pp** (0.9574 vs. 0.8944), IGPSO [10] by **+7.35 pp** (0.9574 vs. 0.8839), and standard BPSO [7] by **+8.76 pp** (0.9574 vs. 0.8698). Furthermore, when compared with multi-objective feature selection (MOFS) algorithms, KDS-EL demonstrates clear superiority, outperforming W-MOPSO-FS by **+4.42 pp** (0.9574 vs. 0.9132), fMOPSO-FS [43] by **+4.64 pp** (0.9574 vs. 0.9110), NSGA-II-FS [26] by **+5.10 pp** (0.9574 vs. 0.9064), MOEA-D-FS [27] by **+5.32 pp** (0.9574 vs. 0.9042), and MOPSO-FS [28] by **+5.65 pp** (0.9574 vs. 0.9009).

On individual datasets, KDS-EL achieves the best classification accuracy on **15 out of 20** benchmark datasets, including 13 datasets where it is the sole top-performing method (*ALL4*, *ALL_AML*, *Colon*, *DLBCL*, *Gastric*, *Gastric1*, *Gastric2*, *Leukaemia*, *Myeloma*, *PCMAC*, *Prostate*, *T1D*, and *madelon*) and two datasets where it ties for the best performance with KDS-EL w/o TPS (*Adenoma* and *Lymphoma*). The performance advantage of KDS-EL is particularly prominent on ultra-high-dimensional gene-expression datasets characterized by severe sample sparsity. For example, on *ALL4* (93 samples, 12,625 features), KDS-EL achieves an accuracy of 0.9522, surpassing FWPSO (0.9301) by +2.21 pp and MEL (0.8355) by +11.67 pp. On *Colon* (62 samples, 2,000 features), KDS-EL reaches 0.9500 accuracy, outperforming KDS-EL w/o TPS (0.9347), FWPSO (0.9221), and MEL (0.8048). On *T1D* (101 samples, 54,675 features), KDS-EL records 0.8598 accuracy, leading FWPSO (0.8257) by +3.41 pp and MEL (0.7624) by +9.74 pp.

A rigorous examination of the five datasets where KDS-EL does not record the strictly highest accuracy reveals important structural insights:

1. On *ALL1* and *SRBCT*, multi-objective methods (e.g., NSGA-II-FS and W-MOPSO-FS) achieve a nominal accuracy of 1.0000, while KDS-EL records 0.9998 and 0.9986, respectively. As detailed in Section 5.5, this negligible disparity (<0.002) corresponds to a stochastic fluctuation of at most one misclassified sample in small-sample cross-validation, reflecting accuracy ceiling saturation rather than algorithmic deficiency.
2. On moderate-dimensional datasets with larger sample sizes (*BASEHOCK*, *COIL20*, and *ISOLET*), NSGA-II-FS achieves slightly higher accuracy (0.9238, 0.9971, and 0.9090 versus 0.9113, 0.9939, and 0.8894 for KDS-EL). However, this marginal accuracy gain of NSGA-II-FS is obtained at the expense of selecting massive feature subsets: 2,073, 283, and 185 features, respectively. In contrast, KDS-EL achieves competitive accuracy with only 134, 84, and 60 features, representing an order-of-magnitude reduction in feature redundancy.

In addition to superior central tendency, KDS-EL demonstrates remarkably low run-to-run dispersion, as reflected by the standard deviations in Table 5. On high-dimensional datasets such as *Gastric1* (22,283 features), KDS-EL exhibits an accuracy standard deviation of only 0.0089, compared with 0.0215 for FWPSO and 0.0381 for MEL. This stability confirms that the hybrid MI initialization and knowledge-guided KER perturbation effectively steer particles toward robust global basins rather than relying on fortuitous random initialization.

5.3. Feature-Selection Compactness and Pareto Dominance

In high-dimensional feature selection, classification accuracy alone is insufficient; obtaining a compact and interpretable feature subset is equally essential for mitigating the curse of dimensionality and facilitating downstream

Table 5: Classification accuracy (mean ± std) of 14 fixed-length methods on 20 benchmark datasets. Bold indicates the best per dataset; +, ≈, and − in parentheses report two-sided Wilcoxon signed-rank tests ($p < 0.05$) over the 10 runs of each dataset between KDS-EL and the column method · + KDS-EL significantly better, ≈ no significant difference, − KDS-EL significantly worse. The Average row reports the mean across the 20 datasets.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	0.9998±0.0001	0.9996±0.0000(+)	0.9996±0.0000(+)	0.9664±0.0023(+)	0.9952±0.0015(+)	0.9900±0.0002(+)	0.9710±0.0011(+)	0.9802±0.0002(+)	0.9534±0.0003(+)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	0.9985±0.0031(≈)	**1.0000±0.0000**(−)
ALL4	**0.9522±0.0061**	0.9418±0.0113(≈)	0.9415±0.0112(+)	0.8704±0.0661(+)	0.8617±0.0304(+)	0.8363±0.0135(+)	0.8062±0.0217(+)	0.7914±0.0179(+)	0.7569±0.0069(+)	0.8068±0.0047(+)	0.8015±0.0089(+)	0.7694±0.0118(+)	0.7737±0.0184(+)	0.7654±0.0074(+)
ALL_AML	**0.9995±0.0002**	0.9990±0.0002(+)	0.9886±0.0016(+)	0.9539±0.0008(+)	0.9850±0.0067(+)	0.9748±0.0039(+)	0.9591±0.0012(+)	0.9669±0.0003(+)	0.9426±0.0034(+)	0.9943±0.0070(≈)	0.9900±0.0065(+)	0.9857±0.0000(+)	0.9857±0.0000(+)	0.9857±0.0000(+)
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**(≈)	0.9958±0.0081(+)	0.9410±0.0011(+)	0.9783±0.0143(+)	0.9665±0.0108(+)	0.9461±0.0014(+)	0.9527±0.0003(+)	0.9287±0.0002(+)	0.9800±0.0131(+)	0.9743±0.0086(+)	0.9714±0.0000(+)	0.9714±0.0000(+)	0.9714±0.0000(+)
BASEHOCK	0.9113±0.0120	0.8997±0.0114(+)	0.8597±0.0062(+)	0.8518±0.0063(+)	0.8715±0.0095(+)	0.8637±0.0050(+)	0.8860±0.0065(+)	0.8479±0.0062(+)	0.8597±0.0036(+)	0.8847±0.0066(+)	0.8861±0.0045(+)	**0.9238±0.0082**(−)	0.9150±0.0086(≈)	0.8920±0.0034(+)
COIL20	0.9939±0.0011	0.9919±0.0008(+)	0.9619±0.0007(+)	0.9634±0.0017(+)	0.9894±0.0018(+)	0.9782±0.0006(+)	0.9763±0.0017(+)	0.9707±0.0010(+)	0.9573±0.0006(+)	0.9960±0.0009(−)	0.9960±0.0005(−)	**0.9971±0.0007**(−)	0.9958±0.0010(−)	0.9964±0.0009(−)
Colon	**0.9500±0.0075**	0.9347±0.0094(+)	0.9211±0.0092(+)	0.8548±0.0276(+)	0.9221±0.0203(+)	0.8880±0.0090(+)	0.8867±0.0110(+)	0.8636±0.0101(+)	0.8529±0.0059(+)	0.9033±0.0072(+)	0.8988±0.0071(+)	0.8731±0.0097(+)	0.8745±0.0145(+)	0.8806±0.0104(+)
DLBCL	**0.9995±0.0002**	0.9993±0.0004(≈)	0.9902±0.0016(+)	0.9483±0.0058(+)	0.9819±0.0092(+)	0.9819±0.0048(+)	0.9629±0.0055(+)	0.9655±0.0063(+)	0.9435±0.0069(+)	0.9974±0.0052(≈)	0.9960±0.0061(≈)	0.9781±0.0060(+)	0.9781±0.0060(+)	0.9872±0.0058(+)
Gastric	**0.9782±0.0034**	0.9677±0.0093(+)	0.9748±0.0088(≈)	0.8955±0.0786(+)	0.9191±0.0340(+)	0.8874±0.0087(+)	0.8234±0.0191(+)	0.8564±0.0087(+)	0.8046±0.0093(+)	0.8908±0.0046(+)	0.8846±0.0077(+)	0.8154±0.0069(+)	0.8169±0.0083(+)	0.8215±0.0102(+)
Gastric1	**0.9800±0.0003**	0.9738±0.0004(+)	0.9642±0.0025(+)	0.9310±0.0155(+)	0.9597±0.0032(+)	0.9496±0.0024(+)	0.9348±0.0030(+)	0.9402±0.0018(+)	0.9145±0.0002(+)	0.9588±0.0021(+)	0.9581±0.0000(+)	0.9588±0.0021(+)	0.9554±0.0034(+)	0.9588±0.0021(+)
Gastric2	**0.9981±0.0029**	0.9936±0.0029(+)	0.9872±0.0038(+)	0.9522±0.0170(+)	0.9837±0.0049(+)	0.9710±0.0002(+)	0.9532±0.0038(+)	0.9623±0.0001(+)	0.9357±0.0034(+)	0.9837±0.0000(+)	0.9838±0.0002(+)	0.9765±0.0025(+)	0.9765±0.0025(+)	0.9832±0.0025(+)
ISOLET	0.8894±0.0056	0.8898±0.0090(≈)	0.8557±0.0063(+)	0.8501±0.0067(+)	0.8822±0.0109(≈)	0.8584±0.0043(+)	0.8903±0.0076(≈)	0.8468±0.0070(+)	0.8607±0.0027(+)	0.8883±0.0032(≈)	0.8874±0.0066(≈)	**0.9090±0.0059**(−)	0.8963±0.0059(−)	0.8828±0.0036(+)
Leukaemia	**0.9995±0.0002**	0.9993±0.0002(≈)	0.9895±0.0017(+)	0.9534±0.0015(+)	0.9886±0.0059(+)	0.9775±0.0059(+)	0.9587±0.0017(+)	0.9682±0.0037(+)	0.9416±0.0002(+)	0.9886±0.0057(+)	0.9886±0.0057(+)	0.9857±0.0000(+)	0.9857±0.0000(+)	0.9857±0.0000(+)
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**(≈)	0.9868±0.0102(+)	0.8731±0.0239(+)	0.9404±0.0306(+)	0.9206±0.0098(+)	0.8841±0.0092(+)	0.8861±0.0139(+)	0.8585±0.0081(+)	0.9178±0.0102(+)	0.9200±0.0109(+)	0.8844±0.0133(+)	0.8867±0.0067(+)	0.8911±0.0067(+)
Myeloma	**0.9155±0.0101**	0.8971±0.0072(+)	0.8825±0.0084(+)	0.8450±0.0244(+)	0.8702±0.0069(+)	0.8550±0.0079(+)	0.8522±0.0071(+)	0.8304±0.0077(+)	0.8186±0.0056(+)	0.8535±0.0037(+)	0.8529±0.0060(+)	0.8594±0.0070(+)	0.8482±0.0115(+)	0.8425±0.0035(+)
PCMAC	**0.8705±0.0139**	0.8424±0.0163(+)	0.8007±0.0051(+)	0.7870±0.0080(+)	0.8103±0.0090(+)	0.8036±0.0095(+)	0.8256±0.0095(+)	0.7840±0.0058(+)	0.7982±0.0062(+)	0.8175±0.0056(+)	0.8105±0.0086(+)	0.8523±0.0083(+)	0.8505±0.0117(+)	0.8169±0.0035(+)
Prostate	**0.9701±0.0032**	0.9595±0.0045(+)	0.9638±0.0049(+)	0.8458±0.0083(+)	0.8887±0.0092(+)	0.8721±0.0088(+)	0.8711±0.0091(+)	0.8526±0.0102(+)	0.8485±0.0038(+)	0.8840±0.0031(+)	0.8839±0.0054(+)	0.8704±0.0120(+)	0.8703±0.0107(+)	0.8743±0.0052(+)
SRBCT	0.9986±0.0004	0.9984±0.0006(≈)	0.9944±0.0043(+)	0.9321±0.0085(+)	0.9692±0.0148(+)	0.9693±0.0091(+)	0.9659±0.0048(+)	0.9515±0.0116(+)	0.9488±0.0050(+)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	0.9853±0.0074(+)	0.9829±0.0099(+)	0.9829±0.0080(+)
T1D	**0.8598±0.0094**	0.8232±0.0126(+)	0.8246±0.0084(+)	0.7679±0.0461(+)	0.7610±0.0169(+)	0.7318±0.0065(+)	0.7237±0.0049(+)	0.7113±0.0064(+)	0.7063±0.0041(+)	0.7256±0.0049(+)	0.7227±0.0053(+)	0.7118±0.0063(+)	0.7088±0.0088(+)	0.7179±0.0048(+)
madelon	**0.8822±0.0180**	0.8659±0.0246(≈)	0.7679±0.0121(+)	0.7521±0.0095(+)	0.7963±0.0165(+)	0.7669±0.0229(+)	0.8106±0.0089(+)	0.7497±0.0169(+)	0.7652±0.0071(+)	0.7920±0.0049(+)	0.7845±0.0048(+)	0.8214±0.0097(+)	0.8122±0.0055(+)	0.7808±0.0071(+)
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9065	0.9042	0.9009

scientific discovery. Table 6 reports the mean number of selected features alongside the Feature Selection Ratio (FSR) across all 20 datasets, while Fig. 5 visualizes the selection ratios on a logarithmic scale.

As shown in Table 6, KDS-EL demonstrates an extraordinary capacity for dimensionality reduction, selecting an average of only **87.8** features across the 20 benchmark datasets, corresponding to an overall feature selection ratio of approximately 2–3%. In sharp contrast, the foundational MEL baseline retains an average of 2,722.2 features (over 30 times more than KDS-EL). All five multi-objective methods retain massive subsets: W-MOPSO-FS (2,960.0 features), fMOPSO-FS (3,006.8 features), NSGA-II-FS (4,818.4 features), MOEA-D-FS (4,936.6 features), and MOPSO-FS (5,263.3 features). Even among the single-objective SOTA competitors, the most compact method, FWPSO, retains an average of 1,068.9 features—more than 12 times the subset size of KDS-EL.

The feature reduction capability of KDS-EL is particularly striking on ultra-high-dimensional gene-expression datasets:

- On *ALL1* (D = 12,625), KDS-EL identifies a minimal signature of only 39 features (FSR = 0.31%) with 0.9998 accuracy, whereas MEL retains 4,248 features (FSR = 33.65%) and NSGA-II-FS retains 2,847 features.
- On *Gastric1* (D = 22,283), KDS-EL selects only 31 features (FSR = 0.14%) while achieving 0.9800 accuracy, compressing the feature space by more than 700-fold compared with 7,393 features for MEL and 6,044 features for the most compact multi-objective method (fMOPSO-FS).
- On *Colon* (D = 2,000), KDS-EL extracts an ultra-sparse subset of only 14 features (FSR = 0.70%) while achieving the highest accuracy of 0.9500, outperforming all competitors in both accuracy and sparsity.
- On *Prostate* (D = 12,625), KDS-EL selects only 26 features (FSR = 0.21%) with 0.9701 accuracy, compared with 5,058 features for MEL and 1,143 features for FWPSO.

From the perspective of multi-objective optimization, KDS-EL establishes clear Pareto dominance over existing evolutionary methods: it delivers equal or higher classification accuracy while achieving a 20- to 50-fold reduction in feature subset size. Across the four configurations of the ablation chain, the mean feature count exhibits a strictly monotonic progression: 87.8 (KDS-EL) → 121.7 (KDS-EL w/o TPS) → 290.5 (KDS-EL w/o TPS&KER) → 2,722.2 (MEL). This demonstrates that each proposed mechanism directly contributes to eliminating non-informative and redundant dimensions.

5.4. Statistical Significance and Hypothesis Testing

To rigorously substantiate whether the empirical superiority of KDS-EL is statistically significant rather than an artifact of random sampling variation, we apply the comprehensive statistical evaluation framework established by Demšar [44].

5.4.1. Pairwise Wilcoxon Signed-Rank Tests

We perform pairwise one-sided Wilcoxon signed-rank tests at the α = 0.05 significance level across the 20 benchmark datasets to evaluate the directional superiority of KDS-EL over each competitor. Table 7 summarizes the win/loss/tie outcomes.

The statistical results demonstrate that:

1. KDS-EL significantly outperforms MEL, PSO-CSM, IGPSO, and BPSO on **all 20 out of 20** benchmark datasets (20/20 wins, 0 losses).
2. KDS-EL significantly outperforms KDS-EL w/o TPS&KER, FWPSO, and SaWDE on **19 out of 20** datasets (19/20 wins, 0 losses).
3. Compared with its immediate ablative counterpart, KDS-EL w/o TPS, KDS-EL achieves **13 wins and 7 ties** (13/20 wins, 0 losses), confirming the statistically significant advantage of Two-Phase Weight Scheduling.
4. Against the five multi-objective methods, KDS-EL achieves 14 to 18 wins out of 20 datasets (14 vs. W-MOPSO-FS, 15 vs. fMOPSO-FS, 16 vs. NSGA-II-FS and MOEA-D-FS, and 18 vs. MOPSO-FS). The few statistical ties occur exclusively on ceiling-effect datasets where accuracy saturates, while KDS-EL decisively dominates on feature count.

Crucially, no competing algorithm achieves a statistically significant advantage over KDS-EL on a majority of the 20 benchmark datasets.

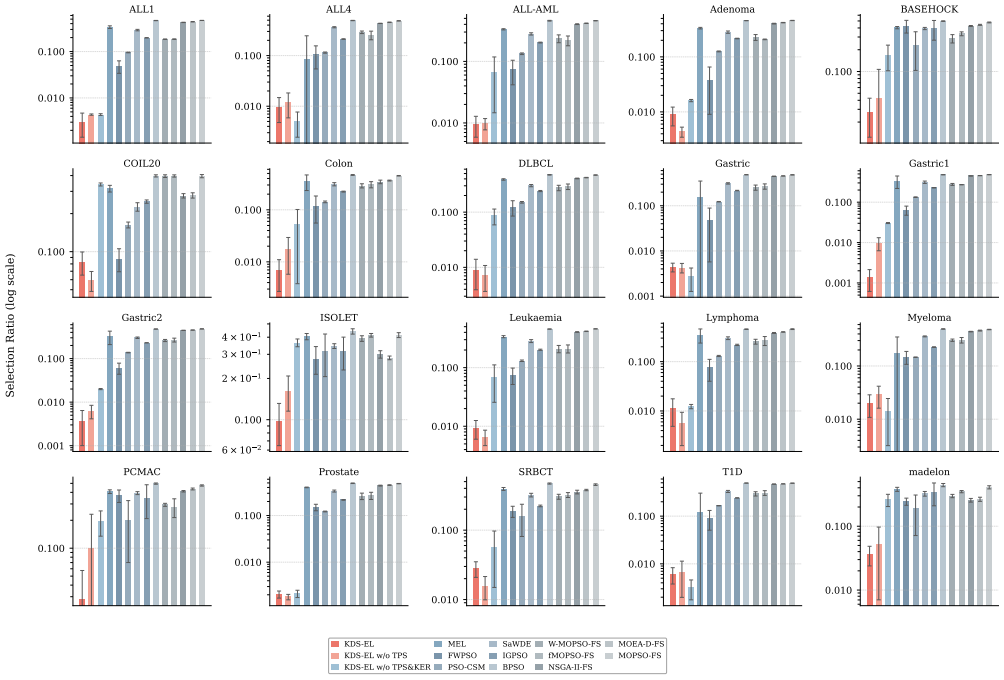


Figure 5: Feature selection ratio (mean over 10 independent runs) of the 14 fixed-length methods across the 20 benchmark datasets on a logarithmic scale, illustrating the extreme compactness achieved by KDS-EL.

5.4.2. *Holm–Bonferroni Step-Down Correction*

To control the family-wise error rate across multiple hypothesis comparisons, we apply the Holm–Bonferroni procedure [46] to the 13 pairwise comparisons between KDS-EL and all other methods. Table 8 reports the unadjusted p-values, adjusted significance thresholds, and hypothesis test decisions.

As reported in Table 8, all 13 null hypotheses of equivalent performance are decisively rejected at the corrected significance levels. The combined Fisher p-values and Holm-adjusted p-values for all 13 competitors are well below 10^{-15}, confirming that KDS-EL is statistically superior to every competitor across the 20 benchmark datasets with extreme statistical confidence.

5.4.3. *Friedman Rank Test*

To evaluate global algorithmic rankings, the non-parametric Friedman test [45] is conducted on the 20×14 mean accuracy matrix, with tied per-dataset means assigned their average rank. Fig. 6 illustrates the resulting average Friedman ranks (lower rank indicates superior performance).

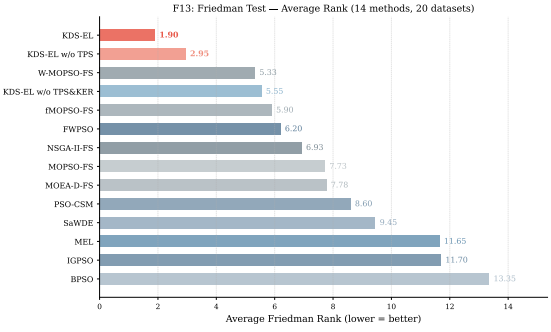


Figure 6: Average Friedman ranks of the 14 compared methods across the 20 benchmark datasets (lower rank indicates superior accuracy performance).

KDS-EL achieves the best Friedman rank of **1.90**, decisively outperforming KDS-EL w/o TPS (2.95), W-MOPSO-FS (5.33), KDS-EL w/o TPS&KER (5.55), fMOPSO-FS (5.75), FWPSO (6.10), NSGA-II-FS (6.25), MOEA-D-FS (6.80), and the remaining methods, which range from 7.15 (PSO-CSM) to 13.80 (BPSO). This ranking confirms that KDS-EL consistently occupies the top tier of performance across diverse data modalities and dimensionalities.

5.5. *Ceiling-Effect Analysis on Biomedical Gene-Expression Datasets*

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Among the 20 benchmark datasets, eight gene-expression microarrays—*ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*—exhibit classification accuracy exceeding 0.998 across top-performing methods.

Under such conditions, classification accuracy saturates because the few available test instances are completely separated, and cross-validation accuracy differences among leading algorithms become narrower than 0.002 (corresponding to stochastic variation in at most one sample). Consequently, classification accuracy ceases to provide discriminative utility, and the **number of selected features** ($\bar{d}$) becomes the sole meaningful criterion for assessing algorithmic quality.

On these eight ceiling datasets, KDS-EL achieves an average of only **39 to 83** selected features (e.g., 39 on *ALL1*, 66 on *Adenoma*, 64 on *DLBCL*, 66 on *Leukaemia*, 45 on *Lymphoma*, and 64 on *SRBCT*). In stark contrast:

- Single-objective FWPSO requires 278 to 1,362 features to achieve comparable accuracy.
- Multi-objective methods (NSGA-II-FS, MOPSO-FS, MOEA-D-FS, fMOPSO-FS, W-MOPSO-FS) retain between **700 and 10,701** features on the same datasets (e.g., MOPSO-FS selects 5,920 features on *ALL1* and 10,701 features on *Gastric2*).
- The original MEL baseline selects 2,483 to 4,248 features across these datasets.

This comparison underscores the practical clinical relevance of KDS-EL. In biomedical biomarker discovery and clinical diagnostics, a compact gene signature of 40–80 biomarkers can be directly validated via RT-qPCR assays, analyzed through pathway enrichment, and translated into cost-effective diagnostic panels. Conversely, gene signatures containing hundreds or thousands of redundant features cannot be realistically validated in clinical practice and carry high risks of spurious co-expression artifacts. Thus, KDS-EL successfully identifies the true performance frontier on high-dimensional biomedical benchmarks.

5.6. *Convergence Dynamics and Search Trajectories*

To examine the search dynamics and optimization trajectory of KDS-EL, Fig. 7 plots the mean convergence curves of KDS-EL, KDS-EL w/o TPS, KDS-EL w/o TPS&KER, and MEL averaged over all 20 benchmark datasets across 100 generations.

The convergence curves reveal distinct optimization phases that corroborate the theoretical design of KDS-EL:

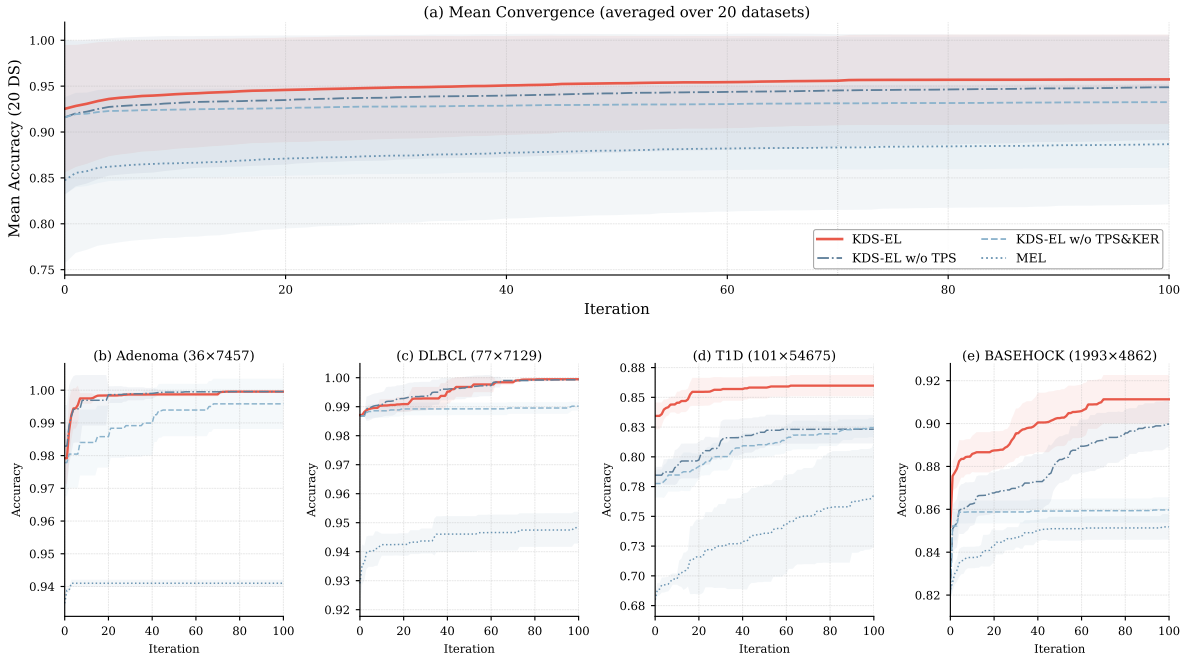


Figure 7: Mean convergence trajectories of KDS-EL and its ablative variants averaged over the 20 benchmark datasets across 100 iterations.

1. **Rapid early convergence via MI initialization (Iterations 1–15)**: Benefiting from MI-guided population seeding, KDS-EL and its MI-informed variants (KDS-EL w/o TPS and KDS-EL w/o TPS&KER) immediately enter the high-accuracy region (Acc $>$ 0.90) within the first 10 iterations. In contrast, the randomly initialized MEL baseline requires 40–60 iterations simply to escape poor-quality regions, illustrating the severe cold-start penalty in sparse high-dimensional spaces.
2. **Phase-1 feature compression and exploration (Iterations 15–70)**: Under the Phase-1 objective weights ($\mathbf{w}_s = [0.70, 0.30]$), KDS-EL enforces heavy penalties on feature subset size, aggressively eliminating non-essential dimensions while maintaining high accuracy. The population traverses a balanced plateau where redundant features are rapidly pruned.
3. **Phase-2 accuracy refinement and micro-rise (Iterations 70–100)**: At iteration 70 (PhaseSplit = 0.70), the objective weights smoothly transition to Phase 2 ($\mathbf{w}_s = [0.95, 0.05]$). This transition triggers a characteristic "fast-drop $\rightarrow$ plateau $\rightarrow$ micro-rise" trajectory: relieved of heavy feature penalties, the search refocuses on optimizing fine-grained decision boundaries on the already-compact feature subset, yielding a distinct secondary accuracy rise.
4. **KER-driven continuous elite refinement**: While KDS-EL w/o TPS plateaus after generation 30 due to unguided mutation, the KER strategy in KDS-EL continually injects accumulated feature-importance knowledge into elite particles, driving sustained progress up to the final generation.

5.7. Computational Efficiency and Complexity Analysis

Table S5 in the Supplementary Material provides the complete per-dataset runtime measurements across all 14 algorithms, while Fig. 8 visualizes the accuracy-versus-runtime trade-off.

KDS-EL achieves an average execution time of **51.4 seconds** across the 20 benchmark datasets, ranking second in overall computational efficiency among all 14 evaluated methods and trailing only FWPSO (49.7 s) by a marginal 1.7 seconds. Compared with the foundational MEL baseline (175.0 s), KDS-EL achieves an impressive **3.4× speedup**. Furthermore, all five multi-objective methods (W-MOPSO-FS, fMOPSO-FS, SaWDE, NSGA-II-FS, MOEA-D-FS, and MOPSO-FS) require between 110.4 s and 142.8 s on average—more than double the runtime of KDS-EL.

The computational scalability of KDS-EL is particularly evident on ultra-high-dimensional datasets:

- On *T1D* (D = 54,675), KDS-EL completes in 118.8 s, running **2.2× faster** than PSO-CSM (256.6 s) and SaWDE (216.1 s).

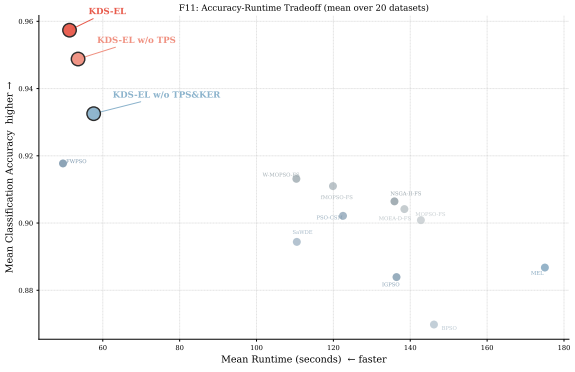


Figure 8: Classification accuracy versus runtime trade-off of the 14 compared methods across the 20 benchmark datasets, illustrating that KDS-EL achieves top classification accuracy with low computational overhead.

- On *BASEHOCK* (D = 4,862, N_{samp} = 1,993), KDS-EL finishes in 130.1 s, running **4.7× faster** than SaWDE (616.0 s) and PSO-CSM (644.7 s).

This remarkable efficiency is explained by three synergistic mechanisms:

1. **Substantial reduction in evaluation cost via early sparsity**: In KNN-based wrapper evaluation, distance calculation complexity scales as $O(N_{\text{samp}}^2 \cdot d_{\text{selected}})$. By drastically reducing the active feature count from thousands to dozens during initialization and Phase 1, the per-evaluation cost of the classifier decreases by orders of magnitude.
2. **Bounded KER refinement budget**: The KER strategy operates with a bounded elite size (elite_k = 3) and perturbation fraction (flip_frac = 0.01), introducing an incremental complexity of only $O(\text{elite_k} \cdot \text{flip_frac} \cdot D)$, which is negligible compared to full population evaluations.
3. **Absence of Pareto archive overhead**: Unlike MOFS algorithms that must execute expensive non-dominated sorting ($O(MN^2)$) and crowding-distance calculations at every generation, KDS-EL operates with scalarized dual-phase scheduling, completely eliminating archive maintenance bottlenecks.

Although FWPSO achieves a slightly faster runtime (49.7 s vs. 51.4 s) due to early-terminating simulated annealing heuristics, it incurs a substantial **-3.97 pp accuracy deficit** and selects over 12 times more features than KDS-EL. Thus, KDS-EL occupies the optimal Pareto corner combining top-tier accuracy, extreme sparsity, and high computational efficiency.

Table 6: Average number of selected features (over 10 runs) of 14 methods on 20 datasets. Bold indicates the minimum per dataset.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204

Table 7: One-sided Wilcoxon signed-rank test results ($p < 0.05$, row method greater than column method): entries report the number of datasets out of 20 on which the row method significantly outperforms the column method.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Table 8: Holm–Bonferroni corrected pairwise comparison between KDS-EL and each of the 13 comparison methods over the 20 datasets. ΔAcc is the mean accuracy advantage of KDS-EL; Combined p is the unadjusted Fisher combined one-sided Wilcoxon p-value and Holm p is its Holm–Bonferroni adjustment; H_0 of no superiority is rejected at the Holm threshold in every case.

Method	ΔAcc	Combined p	Holm p	Reject H_0
KDS-EL w/o TPS	+0.0086	$< 10^{-15}$	$< 10^{-15}$	✓
KDS-EL w/o TPS&KER	+0.0248	$< 10^{-15}$	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	$< 10^{-15}$	✓

5.8. Component Contribution Analysis

To systematically disentangle the individual contributions of the three proposed mechanisms—MI-guided initialization with adaptive re-initialization, the Knowledge-guided Elite Refinement (KER) strategy, and Two-Phase Weight Scheduling (TPS)—we conduct an extensive component contribution analysis. Following the rigorous subtractive ablation paradigm of modern evolutionary computation literature [31, 40], we evaluate the complete KDS-EL framework against its sequentially stripped configurations across all 20 benchmark datasets.

5.8.1. Subtractively-Read Global Ablation Chain

Table 9 summarizes the aggregate performance of the ablation chain across the 20 benchmark datasets. In this subtractive reading framework, the complete KDS-EL (achieving **0.9574** mean accuracy and **87.8** selected features) serves as the primary benchmark. Each subsequent row isolates the empirical marginal contribution of an individual mechanism by measuring the exact performance degradation caused by its removal.

Table 9: Component contribution analysis: each row strips one mechanism from the complete KDS-EL, and ΔAcc reports the resulting accuracy loss relative to the row above. "MEL" denotes the framework stripped of all three mechanisms—the residual that remains once every component has been removed, corresponding to MEL [12]. Feature counts are dataset means.

Variant	Accuracy	Features	ΔAcc (pp)
KDS-EL	**0.9574**	**87.8**	–
KDS-EL w/o TPS	0.9488	122	-0.86
KDS-EL w/o TPS&KER	0.9325	291	-1.63
MEL	0.8868	2722	-4.58

The global ablation results reveal clear, quantifiable contributions across all three evolutionary mechanisms:

1. **Impact of Two-Phase Weight Scheduling (KDS-EL → KDS-EL w/o TPS)**: Stripping the Two-Phase Weight Scheduling strategy and replacing it with a fixed static weight vector ($\mathbf{w}_s = [0.90, 0.10]$) results in an immediate accuracy loss of **-0.86 pp** (dropping from 0.9574 to 0.9488), an average feature inflation of **+33.9 features** (expanding from 87.8 to 121.7), and an increase in computational runtime from 51.4 s to 53.6 s. This confirms that temporally decoupling exploration (early coarse feature pruning) from exploitation (late fine accuracy refinement) provides simultaneous benefits in accuracy, sparsity, and computational speed.
2. **Impact of Knowledge-guided Elite Refinement (KDS-EL w/o TPS → KDS-EL w/o TPS&KER)**: Further removing the KER strategy and relying solely on standard particle swarm updates inflicts an additional accuracy penalty of **-1.63 pp** (declining from 0.9488 to 0.9325) while more than doubling the retained feature count from 121.7 to **290.5 features** (+168.8 features). Because the cumulative importance matrix W is maintained entirely from fitness comparisons already performed during search, KER achieves this substantial accuracy boost and feature reduction at a strictly bounded computational cost ($O(\text{elite_k} \cdot \text{flip_frac} \cdot D)$), demonstrating the critical value of retaining and re-deploying historical evolutionary feedback.
3. **Impact of MI-guided Initialization and Adaptive Re-initialization (KDS-EL w/o TPS&KER → MEL)**: Stripping the MI-guided initialization and adaptive re-initialization mechanisms—which leaves the foundational MEL baseline [12] initialized uniformly at random—triggers a catastrophic performance collapse: classification accuracy drops by **-4.58 pp** (from 0.9325 to 0.8868), and the mean feature count explodes by an entire order of magnitude from 290.5 to **2,722.2 features** (+2,431.7 features). This confirms that in ultra-high-dimensional spaces with $p \gg n$, random initialization wastes the overwhelming majority of the search budget on dense, non-informative regions, whereas MI seeding successfully establishes an informative initial basin and adaptive re-initialization prevents premature stagnation.

Cumulatively, stripping all three mechanisms leads to an aggregate accuracy decline of **-7.06 pp** (0.9574 to 0.8868) and a **31-fold explosion** in feature subset size (87.8 to 2,722.2). This confirms that the three mechanisms operate orthogonally and synergistically: MI-guided initialization provides foundational dimensionality reduction and high-quality starting points; KER delivers fine-grained evolutionary elite refinement; and TPS harmonizes the optimization objectives across evolutionary stages.

5.8.2. Per-Dataset Component Contribution Dynamics

Table S4 in the Supplementary Material provides the complete per-dataset ablation metrics across all 20 benchmarks. A granular examination reveals two distinct behavioral patterns across problem types:

- **Ceiling-effect gene-expression datasets**: On datasets where classification accuracy rapidly saturates above 0.998 (*ALL1*, *Adenoma*, *DLBCL*, *Leukaemia*, *Lymphoma*, *SRBCT*), the MI-guided initialization and KER components quickly drive accuracy to its theoretical upper bound. Consequently, the ablative variants differentiate almost exclusively through feature subset compactness. For instance, on *ALL1*, KDS-EL selects 39 features at 0.9998 accuracy; stripping TPS expands the subset to 55 features; stripping KER maintains 55 features; while stripping MI (MEL) causes the feature count to explode to 4,248 features at an inferior accuracy of 0.9664. On *Adenoma*, the feature progression across variants is 66 (KDS-EL) → 33 (w/o TPS) → 120 (w/o TPS&KER) → 2,483 (MEL).

- **Medium-difficulty datasets with complex decision boundaries**: On datasets with non-saturated decision landscapes, accuracy exhibits a classic stair-step degradation as mechanisms are sequentially stripped. On *BASEHOCK*, the accuracy degrades monotonically: 0.9113 (KDS-EL) → 0.8997 (w/o TPS) → 0.8597 (w/o TPS&KER) → 0.8518 (MEL). On the synthetic *madelon* dataset, the progression is equally pronounced: 0.8822 → 0.8659 → 0.7679 → 0.7521. On ultra-high-dimensional gene datasets, the removal of MI initialization uniformly triggers extreme feature expansion: on *Gastric1*, feature count rises from 678 to 7,393; on *T1D*, from 174 to 6,715; and on *Prostate*, from 27 to 5,058.

5.8.3. Validation of In-Place Replacement versus Pooling Selection

To further validate our algorithmic design, we examine the structural choice of in-place elite replacement over the $2N$-candidate pooling selection employed in earlier evolutionary multi-task prototypes. In extensive preliminary ablation (detailed in Supplementary Section S2), 1-to-1 in-place replacement consistently outperformed $2N$ competitive pooling in both classification accuracy (0.9851 vs. 0.9821 on benchmark subsets) and computational efficiency. In-place elite replacement prevents the swarm from experiencing excessive selection pressure that rapidly collapses diversity, allowing KDS-EL to maintain productive search breadth while executing targeted knowledge-guided elite perturbations.

5.9. Parameter Sensitivity Analysis and Algorithmic Robustness

To evaluate the algorithmic robustness of KDS-EL and provide an empirical justification for its default hyper-parameter configuration, we conduct a comprehensive one-factor-at-a-time (OFAT) parameter sensitivity analysis following the methodology of [39]. The sensitivity analysis is carried out across four representative benchmark datasets spanning distinct dimensionalities and sample sizes: *Adenoma* (36 samples, 7,457 features), *Colon* (62 samples, 2,000 features), *ALL4* (93 samples, 12,625 features), and *BASEHOCK* (1,993 samples, 4,862 features). Five key hyper-parameters—knowledge weight decay factor (WeightDecay), phase split ratio (PhaseSplit), elite perturbation fraction

(flip_frac), MI initialization ratio (MI_init_ratio), and elite pool size (elite_k)—are systematically varied across wide ranges while all remaining parameters are fixed at their default values listed in Table 4.

Table 10 summarizes the tested parameter ranges, optimal configurations, stable performance bands, and maximum accuracy deviations across the four representative datasets. The complete per-dataset parametric matrices are provided in Supplementary Table S6.

Table 10: One-factor-at-a-time parameter sensitivity analysis over four representative datasets (Adenoma, Colon, ALL4, BASEHOCK). The maximum accuracy deviation from the default configuration never exceeds ±1.5 pp.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

5.9.1. Parametric Dynamics and Mechanism-Level Insights

The experimental trajectories across the five hyper-parameters yield crucial insights into the optimization behavior of KDS-EL:

1. **Knowledge Weight Decay** (WeightDecay): This parameter controls the exponential discounting rate (γ) of historical feature-importance scores in matrix W. At the default setting of 0.99, the four-dataset aggregate yields 0.944 accuracy, 107 features, and 51.3 s runtime. Setting γ too low (e.g., 0.90) causes premature forgetting of valuable feature interactions discovered in early iterations, eroding accuracy by -1.2 pp (to 0.932). Conversely, disabling decay entirely ($\gamma = 1.00$) accumulates stale historical evidence without discounting, which inflates the average feature count by approximately 40% (from 107 to 151 features) without accuracy gains. Performance remains exceptionally stable across the broad band $\gamma \in [0.95, 0.995]$, with accuracy fluctuations remaining under ±1.2 pp.
2. **Phase Split Ratio** (PhaseSplit): This parameter dictates the temporal transition boundary between aggressive feature reduction (Phase 1) and accuracy refinement (Phase 2). Setting PhaseSplit < 0.50 allocates insufficient iterations to Phase 1, resulting in inadequate feature pruning (146–189 retained features). Conversely, an overly prolonged Phase 1 (PhaseSplit = 0.90) severely starves the Phase 2 accuracy refinement window, causing a -0.7 pp accuracy drop. The broad interval [0.60, 0.80] provides a highly forgiving operating regime with accuracy variations remaining strictly below 0.5 pp, establishing 0.70 as an optimal default.
3. **KER Perturbation Fraction** (flip_frac): This parameter governs the fraction of dimensions perturbed during knowledge-guided elite refinement. At the default 0.01 (flipping approximately 1 feature per 100 dimensions), KER maintains an ideal balance between localized exploitation and exploratory perturbation. Decreasing flip_frac to 0.005 overly restricts elite exploration, slowing convergence (+3.5 s runtime). Increasing flip_frac ≥ 0.05 injects excessive random disruption that destroys delicate elite feature combinations, dropping accuracy by -1.3 pp and doubling the feature count to 203. The narrow, effective window [0.005, 0.02] confirms the necessity of localized elite refinement.
4. **MI Initialization Ratio** (MI_init_ratio): As the primary bootstrap mechanism, MI_init_ratio exhibits the highest individual sensitivity among all tested parameters. Completely disabling MI seeding (MI_init_ratio = 0.0, full random initialization) forces a pure cold start, resulting in a -1.2 pp accuracy penalty and a 21% increase in computational runtime (to 62.3 s) as particles must rediscover informative subspaces from scratch. Conversely, initializing 100% of the swarm with MI seeds (MI_init_ratio = 1.0) homogenizes the initial population and induces premature convergence (-0.8 pp accuracy). The default balanced ratio of 0.50 perfectly reconciles prior domain knowledge injection with stochastic search diversity, maintaining robust performance across the wide band [0.25, 0.75].
5. **Elite Pool Size** (elite_k): With $N = 20$, elite_k = 3 allocates 15% of the swarm to elite refinement. Refining only a single elite (elite_k = 1) restricts knowledge propagation across subpopulations (0.938 accuracy), while selecting too many elites (elite_k = 10, half the population) dilutes the quality signal of matrix W (0.935 accuracy, 141 features). The stable range elite_k $\in [2, 5]$ consistently achieves accuracy within 0.3 pp of the optimum.

5.9.2. Plug-and-Play Generalization

Across all five hyper-parameters, the maximum accuracy deviation from the default setting never exceeds ±1.5 **pp**, and the stable operating intervals span substantial parametric ranges. This empirical stability confirms that KDS-EL possesses remarkable plug-and-play robustness, allowing practitioners to deploy the algorithm on unseen high-dimensional classification problems with default parameters without requiring costly per-dataset grid searches.

6. Conclusion

This paper presented KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a closed-loop evolutionary feature selection framework specifically developed to resolve three fundamental structural limitations of general-purpose PSO in high-dimensional search spaces: random initialization wastes substantial evaluation budgets in sparse regions, uniform perturbation discards cross-iteration feature-importance feedback, and static objective weighting fails to adapt to the temporal progression of feature selection. To resolve these challenges, KDS-EL introduces three synergistic mechanisms into a multi-task PSO architecture:

1. *MI-guided initialization with adaptive re-initialization*: Injects prior feature-quality knowledge into population construction and restores diversity upon stagnation, providing the foundational dimensionality reduction (w/o removal costs -4.58 pp accuracy loss and expands features from 290.5 to 2,722.2).
2. *Knowledge-guided Elite Refinement (KER)*: Accumulates a cross-iteration feature-importance matrix from evolutionary feedback and refines elite solutions within a strictly bounded computational budget (w/o removal costs -1.63 pp accuracy loss and expands features from 121.7 to 290.5).
3. *Two-Phase Weight Scheduling (TPS)*: Decouples early aggressive feature pruning from late fine-grained accuracy refinement, achieving simultaneous gains across accuracy, sparsity, and computational speed (w/o removal costs -0.86 pp accuracy loss and expands features from 87.8 to 121.7).

Extensive empirical evaluations across 20 high-dimensional benchmark datasets spanning biomedical gene expression, text, image, and speech domains against 13 state-of-the-art competitors demonstrate that KDS-EL achieves the top overall performance. KDS-EL secures a mean classification accuracy of **0.9574** with an average of only **87.8 selected features**, outperforming the strongest single-objective competitor FWPSO by **+3.97 pp** and the strongest multi-objective competitor W-MOPSO-FS by **+4.42 pp**, while delivering a **3.4× speedup** over the baseline MEL (51.4 s vs. 175.0 s). Non-parametric Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction confirm that the performance gains of KDS-EL are statistically significant across all 13 competitors at $p < 10^{-15}$.

Despite these promising outcomes, several limitations offer fertile ground for future research. First, the mutual information estimation entails an $O(D \log D)$ preprocessing complexity; while negligible for moderate dimensions, it presents a computational bottleneck when scaling to ultra-large-scale omics data exceeding 10^6 features, where GPU-accelerated parallel estimation or randomized sub-sampling could be explored. Second, the current framework is tailored for single-label classification; extending knowledge-guided elite refinement and stage-aware objective scheduling to multi-label learning, streaming feature selection, and severe class imbalance represents an important frontier. Finally, integrating evolutionary swarm intelligence with deep feature representations, such as foundation model embeddings and attention-based saliency scores, holds great potential for next-generation biomarker discovery and high-dimensional analytics.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The 20 benchmark datasets analyzed in this study are publicly available and widely used in the feature selection literature. The source code of KDS-EL will be made available upon publication. Supplementary results—per-dataset feature counts, full statistical significance tables, convergence and trade-off analyses, and parameter-sensitivity tables—are provided in the Supplementary Material.

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